

Microduck Remaining Scenarios - End-to-End Evidence

- Microduck Remaining Scenarios: End-to-End Evidence
 - 1. Executive status
 - 2. Evidence boundaries
 - 3. Historical attempts discovered
 - 4. Repository-wide validation context
 - 5. Running: `running-v4`
 - 6. Stilt Walking: `stilts-v3`
 - 7. Swing: `swing-v3`
 - 8. Historical Running: `running-v1`
 - 9. Historical Running: `running-v2`
 - 10. Historical Running: `running-v3`
 - 11. Historical Stilt Walking: `stilts-v1`
 - 12. Historical Stilt Walking: `stilts-v2`
 - 13. Historical Swing: `swing-v1`
 - 14. Historical Swing: `swing-v2`
 - 15. Stilt training failure and ELU diagnosis
 - 16. Tracked reproduction recipes
 - 17. Interpretation
 - 18. Final conclusion

Workflow dates: September 7-8, 2026
Evidence root: `rlx/artifacts/scenarios-e2e-20260907`
Report result: **PASS**

1. Executive status

This report is generated from saved JSON and artifacts. It does not infer a successful skill from training completion, reward magnitude, finite output, an exported ONNX file, or a playable video. A selected scenario is complete only when its API jobs, independent audit, video validation, and matching current artifact hashes all pass.

Scenario	Selected attempt	Skill audit	Held-out controls	Video	E2E
Running	<code>running-v4</code>	PASS	5/5	PASS	complete
Stilt Walking	<code>stilts-v3</code>	PASS	5/5	PASS	complete
Swing	<code>swing-v3</code>	PASS	5/5	PASS	complete

All selected evidence chains pass under the stated nominal simulation conditions. Swing is teacher-initialized PPO; repeated nominal Swing seeds are not independent robustness trials.

2. Evidence boundaries

- Training and deterministic audit are separate. A recipe may use assisted Swing resets, but evaluation and audit start at rest.
- The effective audit environment uses the XML actuator model only. It is nominal local MuJoCo evidence, not BAM, mjlabs GPU, hardware, or sim-to-real certification.
- Stilt morphology is part of the experiment identity. The selected recipes request 2 cm morphology; reports must not substitute the API default of 10 cm.
- Fixed locomotion commands are taken from the selected recipe and reported as actual numeric values. They are not reconstructed from reward or displacement.
- The built-in recipe metrics wrapper captures the active locomotion command before `env.step()` and attaches physical metrics after the transition. This fixes the command/metric boundary when commands resample on the step boundary.
- Locomotion evaluator version 2 anchors intended heading at each twist-command change and advances it only by commanded yaw. Signed displacement therefore measures net progress along the intended command path; a closed circle cannot pass by accumulating path length while returning to its start.
- API outcome handling accepts an explicitly failed skill evaluation long enough to collect render/export evidence. A later successful render or export does not change the skill verdict.
- Outcome inference is deliberately limited: without `audit.json`, no skill verdict exists. API JSON and logs may establish TRAIN FAILED, TRAINING ACTIVE, TRAINED / UNAUDITED, or NOT RUN, but none is a pass. No pass is inferred from filenames, checkpoint existence, finite telemetry, HTTP success, screenshots, or videos.

3. Historical attempts discovered

Scenario	Attempt	Evidence status	Evidence
running	running-v1	RECIPE ONLY	running-v1.json
running	running-v2	FAIL	audit.json
running	running-v3	FAIL	audit.json
running	running-v4	PASS	audit.json
stilts	stilts-v1	TRAIN FAILED at 1,980,416/4,001,792	stilts-v1-api.json
stilts	stilts-v2	FAIL	audit.json
stilts	stilts-v3	PASS	audit.json
swing	swing-v1	FAIL	audit.json
swing	swing-v2	FAIL	audit.json
swing	swing-v3	PASS	audit.json

Only directories containing `audit.json` carry a skill verdict.

4. Repository-wide validation context

These are recorded validation results from commands run during this workflow. Known broader-suite limitations remain separate from the native acceptance checks:

Check	Saved result	Log
RLX native suite	418 passed, 2 skipped	rlx-native-tests-final.log
Broader microduck_local suite	12 failed, 384 passed, 1 skipped	microduck-contract-tests.log
Full RLX collection	collection blocked: optional <code>mjlab</code> package missing	rlx-tests.log
Viewer Node tests	74 / 74 / 0	node-tests.log
Viewer build	Compiled successfully	node-build.log
Viewer lint	eslint	node-lint.log
Report regression tests	Ran 18 tests / OK	report-tests.log
Python static correctness	All checks passed!	ruff.log
Studio UI playback	"passed":true	studio-ui-validation.log

- The full RLX collection is blocked because the optional `mjlab` package is not installed in the captured environment. Native non-`mjlab` RLX tests passed separately.
- The broader `microduck_local` run has 12 existing failures: 11 BAM/step pre-optimization golden mismatches and one symmetry drift assertion. These failures are not hidden or reclassified by this report.

5. Running: `running-v4`

Scenario status: PASS Complete API/audit/video evidence chain: verified

Recipe and effective audit environment

The exact requested recipe follows. These values describe training; they do not override the deterministic audit environment.

```
{
  "actionDelay": false,
  "checkpointInterval": 524288,
  "clipCoefficient": 0.2,
  "domainRand": false,
  "entropyCoefficient": 0,
  "evalSteps": 600,
  "experimentId": "running",
  "gamma": 0.99,
  "initialStd": 0.6,
  "learningRate": 0.0001,
  "locomotionForwardCommand": 0.75,
  "maxEpisodeS": 12,
```

```

"maxGradNorm": 0.5,
"normalizeRewards": true,
"numEnvs": 16,
"numMinibatches": 4,
"numSteps": 128,
"obsNoise": false,
"profile": "full",
"randomYaw": false,
"renderSeconds": 12,
"resumeFromCheckpoint": true,
"rewardWeights": {
  "air_time": 1,
  "flight": 4,
  "foot_clearance": 0.5,
  "head_up": 1,
  "keep_pace": 12,
  "plant_the_foot": 0.05,
  "pose": 0.1,
  "smooth_moves": 0.1,
  "stay_upright": 2,
  "track_turn": 2,
  "yaw_tracking": 8
},
"runName": "running-e2e-20260907-v4",
"seed": 7,
"totalTimesteps": 2097152,
"updateEpochs": 4
}

```

Audit setting	Actual value
action_delay	no
actuator	xml
domain_rand	no
locomotion_forward_command	0.7500
max_episode_s	12
obs_noise	no
random_yaw	no
stilt_blend	0
stilt_height_cm	2
stilt_mass_kg	n/a
swing_initial_angle_deg	0
swing_initial_rate_rad_s	0

Audit setting	Actual value
swing_planar_actions	no
weight_overrides	{'keep_pace': 12, 'track_turn': 2, 'flight': 4, 'air_time': 1, 'stay_upright': 2, 'pose': 0.1, 'head_up': 1, 'foot_clearance': 0.5, 'smooth_moves': 0.1, 'plant_the_foot': 0.05, 'yaw_tracking': 8}

The requested fixed forward command is **0.7500 m/s**. If this value is *n/a*, the environment's native command sequence is evaluated rather than a fixed command.

Per-episode skill evidence

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
501	0	600	yes	1.0000	1.0000	0.7622	6.2711	1.0162	PASS
502	0	600	yes	1.0000	1.0000	0.7746	6.4911	1.0329	PASS
503	0	600	yes	1.0000	1.0000	0.7383	3.8483	0.9844	PASS
504	0	600	yes	1.0000	1.0000	0.7600	5.6116	1.0134	PASS
505	0	600	yes	1.0000	1.0000	0.7644	9.0130	1.0192	PASS

Per-episode physics details

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
501	0	0.3050 / 0.4550	0.6950 / 0.5450	0.3867	167	85 / 85	85 / 85
502	0	0.3117 / 0.4367	0.6883 / 0.5633	0.3983	168	85 / 85	86 / 85
503	0	0.3167 / 0.4367	0.6833 / 0.5633	0.3900	169	85 / 85	85 / 85
504	0	0.3583 / 0.4333	0.6417 / 0.5667	0.3500	170	85 / 85	86 / 85
505	0	0.3050 / 0.4400	0.6950 / 0.5600	0.3983	167	85 / 85	85 / 85

Per-episode failures

- **seed 501, episode 0:** none
- **seed 502, episode 0:** none
- **seed 503, episode 0:** none
- **seed 504, episode 0:** none
- **seed 505, episode 0:** none

Controls

Zero control

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
501	0	49	no	0.7347	1.0000	0.1267	0.1165	0.1689	FAIL
502	0	44	no	0.7045	1.0000	0.1309	0.1074	0.1745	FAIL
503	0	46	no	0.7174	1.0000	0.1280	0.1098	0.1707	FAIL
504	0	48	no	0.7292	1.0000	0.1254	0.1126	0.1672	FAIL
505	0	47	no	0.7234	1.0000	0.1278	0.1123	0.1704	FAIL

Physical details:

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
501	0	0.9388 / 0.9592	0.0612 / 0.0408	0.0408	0	2 / 1	2 / 2
502	0	0.9773 / 0.9773	0.0227 / 0.0227	0.0227	0	0 / 0	1 / 1
503	0	0.9783 / 0.9783	0.0217 / 0.0217	0.0217	0	0 / 0	1 / 1
504	0	0.9583 / 0.9167	0.0417 / 0.0833	0.0417	0	1 / 1	1 / 1
505	0	0.9787 / 0.9574	0.0213 / 0.0426	0.0213	0	0 / 1	1 / 1

Failures:

- **seed 501, episode 0:** command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 502, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 503, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

- **seed 504, episode 0:** command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 505, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

Random control

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
501	0	45	no	0.7778	1.0000	0.1358	0.1149	0.1811	FAIL
502	0	44	no	0.7273	1.0000	0.1366	0.1180	0.1822	FAIL
503	0	43	no	0.7442	1.0000	0.1426	0.1169	0.1901	FAIL
504	0	44	no	0.7500	1.0000	0.1444	0.1214	0.1926	FAIL
505	0	45	no	0.7778	1.0000	0.1329	0.1125	0.1773	FAIL

Physical details:

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
501	0	0.9778 / 0.9111	0.0222 / 0.0889	0.0222	0	0 / 1	1 / 1
502	0	0.9545 / 0.9091	0.0455 / 0.0909	0.0227	1	1 / 1	1 / 2
503	0	0.9767 / 0.9302	0.0233 / 0.0698	0.0233	0	0 / 1	1 / 2
504	0	0.9773 / 0.9318	0.0227 / 0.0682	0.0227	0	0 / 1	1 / 1
505	0	0.9778 / 0.9556	0.0222 / 0.0444	0.0222	0	0 / 1	1 / 1

Failures:

- **seed 501, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 502, episode 0:** aerial fraction below running target; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

- **seed 503, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 504, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 505, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

Initial pretrained baseline

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
501	0	600	yes	1.0000	1.0000	0.6664	-0.2392	0.8886	FAIL

Physical details:

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
501	0	0.3733 / 0.4867	0.6267 / 0.5133	0.275 0	156	78 / 78	79 / 78

Failures:

- **seed 501, episode 0:** command-directed displacement below target

Physics metric means

Control	Seed	Metric	Mean
trained	501	command_forward_m_s	0.7500
trained	501	command_lateral_m_s	0.0000
trained	501	command_yaw_rad_s	0.0000
trained	501	forward_speed_m_s	0.7622
trained	501	heading_forward_speed_m_s	0.7622
trained	501	heading_forward_x	0.7056
trained	501	heading_forward_y	-0.6259
trained	501	heading_lateral_speed_m_s	-0.0108
trained	501	left_foot_contact	0.3050

Control	Seed	Metric	Mean
trained	501	right_foot_contact	0.4550
trained	501	upright	0.9990
trained	501	world_x_m	3.5580
trained	501	world_y_m	-2.2906
trained	502	command_forward_m_s	0.7500
trained	502	command_lateral_m_s	0.0000
trained	502	command_yaw_rad_s	0.0000
trained	502	forward_speed_m_s	0.7746
trained	502	heading_forward_speed_m_s	0.7746
trained	502	heading_forward_x	0.6966
trained	502	heading_forward_y	-0.6845
trained	502	heading_lateral_speed_m_s	0.0150
trained	502	left_foot_contact	0.3117
trained	502	right_foot_contact	0.4367
trained	502	upright	0.9987
trained	502	world_x_m	3.4742
trained	502	world_y_m	-2.7409
trained	503	command_forward_m_s	0.7500
trained	503	command_lateral_m_s	0.0000
trained	503	command_yaw_rad_s	0.0000
trained	503	forward_speed_m_s	0.7383
trained	503	heading_forward_speed_m_s	0.7383
trained	503	heading_forward_x	0.4253
trained	503	heading_forward_y	-0.7713
trained	503	heading_lateral_speed_m_s	0.0190
trained	503	left_foot_contact	0.3167
trained	503	right_foot_contact	0.4367
trained	503	upright	0.9991
trained	503	world_x_m	2.8724
trained	503	world_y_m	-2.7943
trained	504	command_forward_m_s	0.7500
trained	504	command_lateral_m_s	0.0000
trained	504	command_yaw_rad_s	0.0000

Control	Seed	Metric	Mean
trained	504	forward_speed_m_s	0.7600
trained	504	heading_forward_speed_m_s	0.7600
trained	504	heading_forward_x	0.6176
trained	504	heading_forward_y	-0.6863
trained	504	heading_lateral_speed_m_s	0.0110
trained	504	left_foot_contact	0.3583
trained	504	right_foot_contact	0.4333
trained	504	upright	0.9991
trained	504	world_x_m	3.4001
trained	504	world_y_m	-2.4661
trained	505	command_forward_m_s	0.7500
trained	505	command_lateral_m_s	0.0000
trained	505	command_yaw_rad_s	0.0000
trained	505	forward_speed_m_s	0.7644
trained	505	heading_forward_speed_m_s	0.7644
trained	505	heading_forward_x	0.9912
trained	505	heading_forward_y	0.1069
trained	505	heading_lateral_speed_m_s	0.0404
trained	505	left_foot_contact	0.3050
trained	505	right_foot_contact	0.4400
trained	505	upright	0.9992
trained	505	world_x_m	4.3959
trained	505	world_y_m	0.6080
zero	501	command_forward_m_s	0.7500
zero	501	command_lateral_m_s	0.0000
zero	501	command_yaw_rad_s	0.0000
zero	501	forward_speed_m_s	0.1267
zero	501	heading_forward_speed_m_s	0.1267
zero	501	heading_forward_x	0.9999
zero	501	heading_forward_y	0.0143
zero	501	heading_lateral_speed_m_s	-0.0002
zero	501	left_foot_contact	0.9388
zero	501	right_foot_contact	0.9592

Control	Seed	Metric	Mean
zero	501	upright	0.8967
zero	501	world_x_m	0.0236
zero	501	world_y_m	0.0006
zero	502	command_forward_m_s	0.7500
zero	502	command_lateral_m_s	0.0000
zero	502	command_yaw_rad_s	0.0000
zero	502	forward_speed_m_s	0.1309
zero	502	heading_forward_speed_m_s	0.1309
zero	502	heading_forward_x	0.9998
zero	502	heading_forward_y	0.0176
zero	502	heading_lateral_speed_m_s	-0.0009
zero	502	left_foot_contact	0.9773
zero	502	right_foot_contact	0.9773
zero	502	upright	0.8934
zero	502	world_x_m	0.0220
zero	502	world_y_m	-0.0005
zero	503	command_forward_m_s	0.7500
zero	503	command_lateral_m_s	0.0000
zero	503	command_yaw_rad_s	0.0000
zero	503	forward_speed_m_s	0.1280
zero	503	heading_forward_speed_m_s	0.1280
zero	503	heading_forward_x	1.0000
zero	503	heading_forward_y	-0.0024
zero	503	heading_lateral_speed_m_s	-0.0012
zero	503	left_foot_contact	0.9783
zero	503	right_foot_contact	0.9783
zero	503	upright	0.8947
zero	503	world_x_m	0.0211
zero	503	world_y_m	-0.0012
zero	504	command_forward_m_s	0.7500
zero	504	command_lateral_m_s	0.0000
zero	504	command_yaw_rad_s	0.0000
zero	504	forward_speed_m_s	0.1254

Control	Seed	Metric	Mean
zero	504	heading_forward_speed_m_s	0.1254
zero	504	heading_forward_x	0.9999
zero	504	heading_forward_y	0.0114
zero	504	heading_lateral_speed_m_s	-0.0009
zero	504	left_foot_contact	0.9583
zero	504	right_foot_contact	0.9167
zero	504	upright	0.8968
zero	504	world_x_m	0.0223
zero	504	world_y_m	-0.0003
zero	505	command_forward_m_s	0.7500
zero	505	command_lateral_m_s	0.0000
zero	505	command_yaw_rad_s	0.0000
zero	505	forward_speed_m_s	0.1278
zero	505	heading_forward_speed_m_s	0.1278
zero	505	heading_forward_x	0.9998
zero	505	heading_forward_y	-0.0199
zero	505	heading_lateral_speed_m_s	0.0012
zero	505	left_foot_contact	0.9787
zero	505	right_foot_contact	0.9574
zero	505	upright	0.8998
zero	505	world_x_m	0.0249
zero	505	world_y_m	0.0008
initial	501	command_forward_m_s	0.7500
initial	501	command_lateral_m_s	0.0000
initial	501	command_yaw_rad_s	0.0000
initial	501	forward_speed_m_s	0.6664
initial	501	heading_forward_speed_m_s	0.6664
initial	501	heading_forward_x	0.0081
initial	501	heading_forward_y	0.0915
initial	501	heading_lateral_speed_m_s	0.0041
initial	501	left_foot_contact	0.3733
initial	501	right_foot_contact	0.4867
initial	501	upright	0.9989

Control	Seed	Metric	Mean
initial	501	world_x_m	0.4349
initial	501	world_y_m	1.8893

PPO transitions and losses

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	1,024
Optimizer minibatches	16,384
Observed transitions	2,097,152
Requested transitions	2,097,152
Monotonic update steps	yes
Training evidence gate	PASS
Maximum approximate KL	2.8238
Last-100 mean_loss	-0.0225
Last-100 policy_loss	-0.0229
Last-100 value_loss	0.0003
Last-100 entropy	-0.2174
Last-100 approximate_kl	0.0308
Last-100 clip_fraction	0.3393
Last-100 explained_variance	0.5853

Transitions	Steps	Raw return	Skill	Failures
0	600	9,835.52	FAIL	command-directed displacement below target
524,288	600	11,023.29	PASS	none
1,048,576	600	11,253.39	PASS	none
1,572,864	600	11,617.39	PASS	none
2,097,152	600	11,467.54	PASS	none

Initialization and PPO contribution

Initialization evidence	Value
Source checkpoint	/Volumes/ExternalSSD/geoagent/microduck-lab/rlx/runs/studio/running/running-e2e-20260907-v4/running.safetensors
Source SHA-256	c7232327f930cd933eca57fa6454d992ee56976a785db8ca0c5913910eb52b8b
Teacher-assisted	n/a

Initialization evidence	Value
Initial policy role	pretrained baseline, not a null control
Comparison seed	501
PPO raw-return change vs initialization	1,683.18

This run continues a pretrained policy. Its initial skill is not credited to PPO. A nonzero parameter change proves optimization occurred, not that it improved the initializer. Compare the baseline and final physical outcomes and raw returns before attributing any benefit to PPO.

ONNX, controls, and source identity

Check	Value
Policy SHA-256	f48dc18840a428c66dbca23d83a1f36a82a6657e88b7450acd7e7c23d7cc5745
ONNX input	batch x 61
ONNX output	batch x 14
ONNX contract	PASS
Native/ONNX max absolute error	1.431e-06
Parameter L2 change	7.6725
Negative controls failed as required	yes
Audit scope	Nominal local simulation only; no hardware certification. No assisted Swing starts.

API workflow

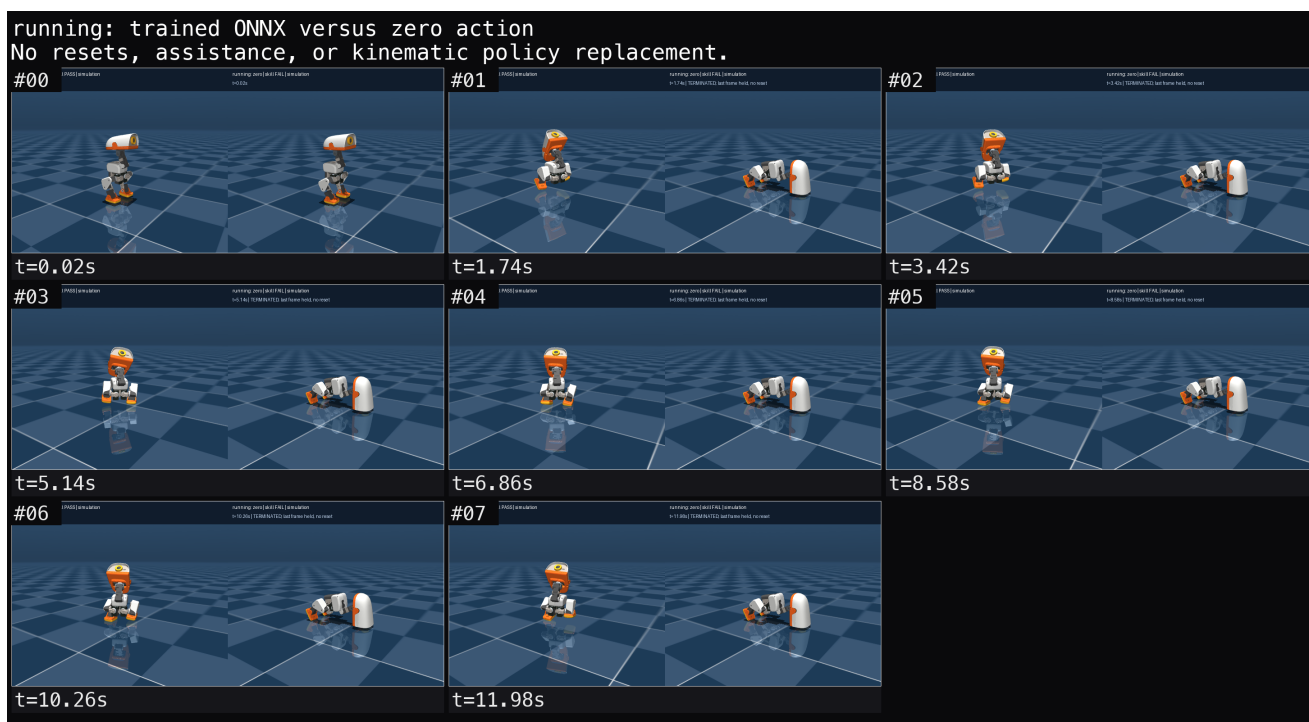
Action	Accepted	Phase	Exit code	Finished
train	yes	succeeded	0	2026-09-08T01:30:42.069Z
eval	yes	succeeded	0	2026-09-08T01:30:57.575Z
render	yes	succeeded	0	2026-09-08T01:31:09.139Z
export	yes	succeeded	0	2026-09-08T01:31:14.888Z

Video and graph evidence

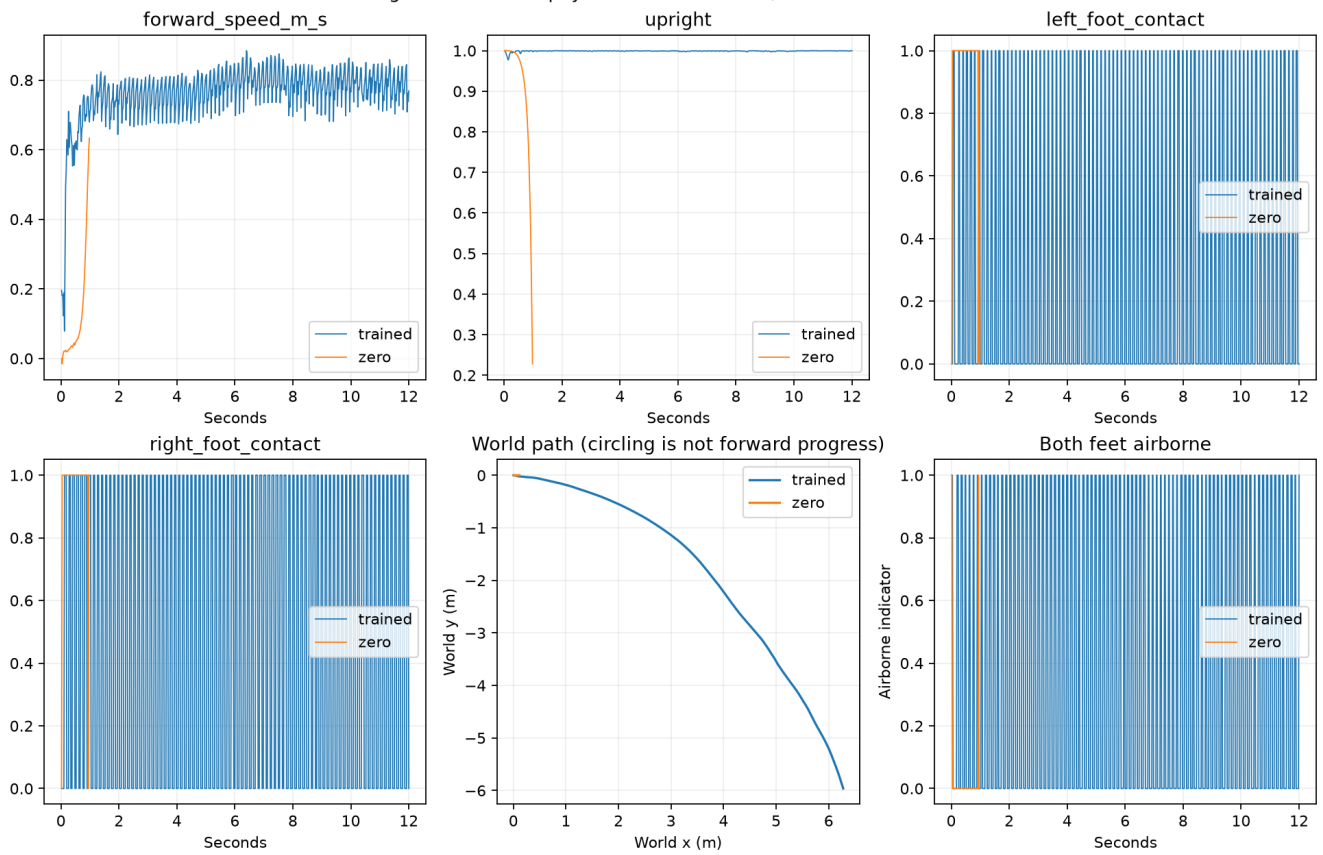
Video validation	Value
Validation status	PASS
API bytes equal local render	yes
HTTP byte range	yes
Browser playback	{"currentTime": 6, "duration": 12, "error": null, "height": 360, "width": 640}

File	Codec	Dimensions	FPS	Frames	Seconds	SHA-256
api-video.mp4	h264	640 x 360	25/ 1	300	12.00000 0	cf1b3fc6abd932ae0693451993ff3fcf0f081a8046ab42c68e0f211966e23655
comparison.mp4	h264	960 x 412	25/ 1	300	12.00000 0	522aa57a3cbdebb95650bbfd312cc491fca44753c0992f1fee50e90754dfbb18

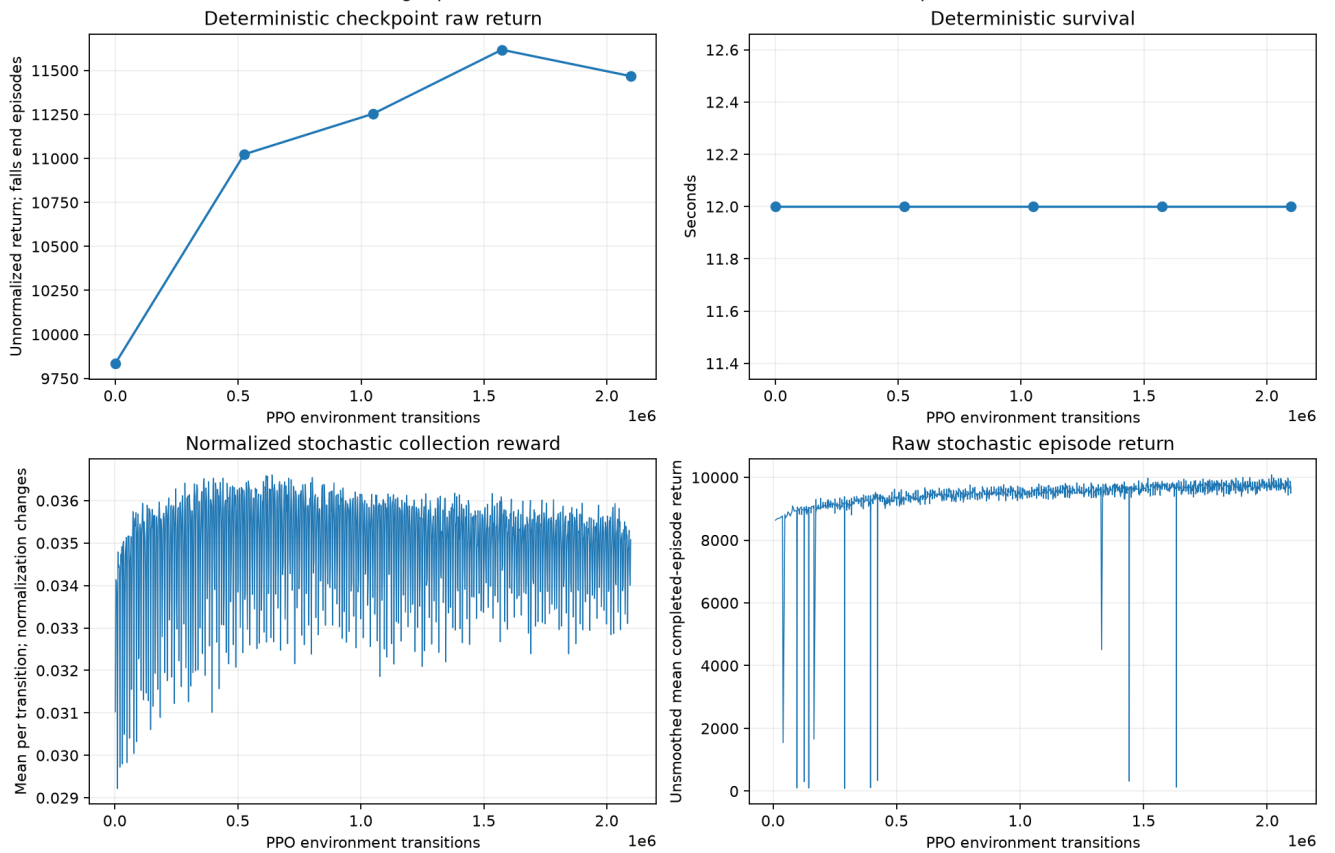
- comparison_sheet.png
- physical-tracking.png
- reward-learning.png
- ppo-losses.png
- browser-video.png
- studio-evaluation.png
- comparison.mp4
- api-video.mp4
- audit.json
- video-validation.json
- running-v4-api.json



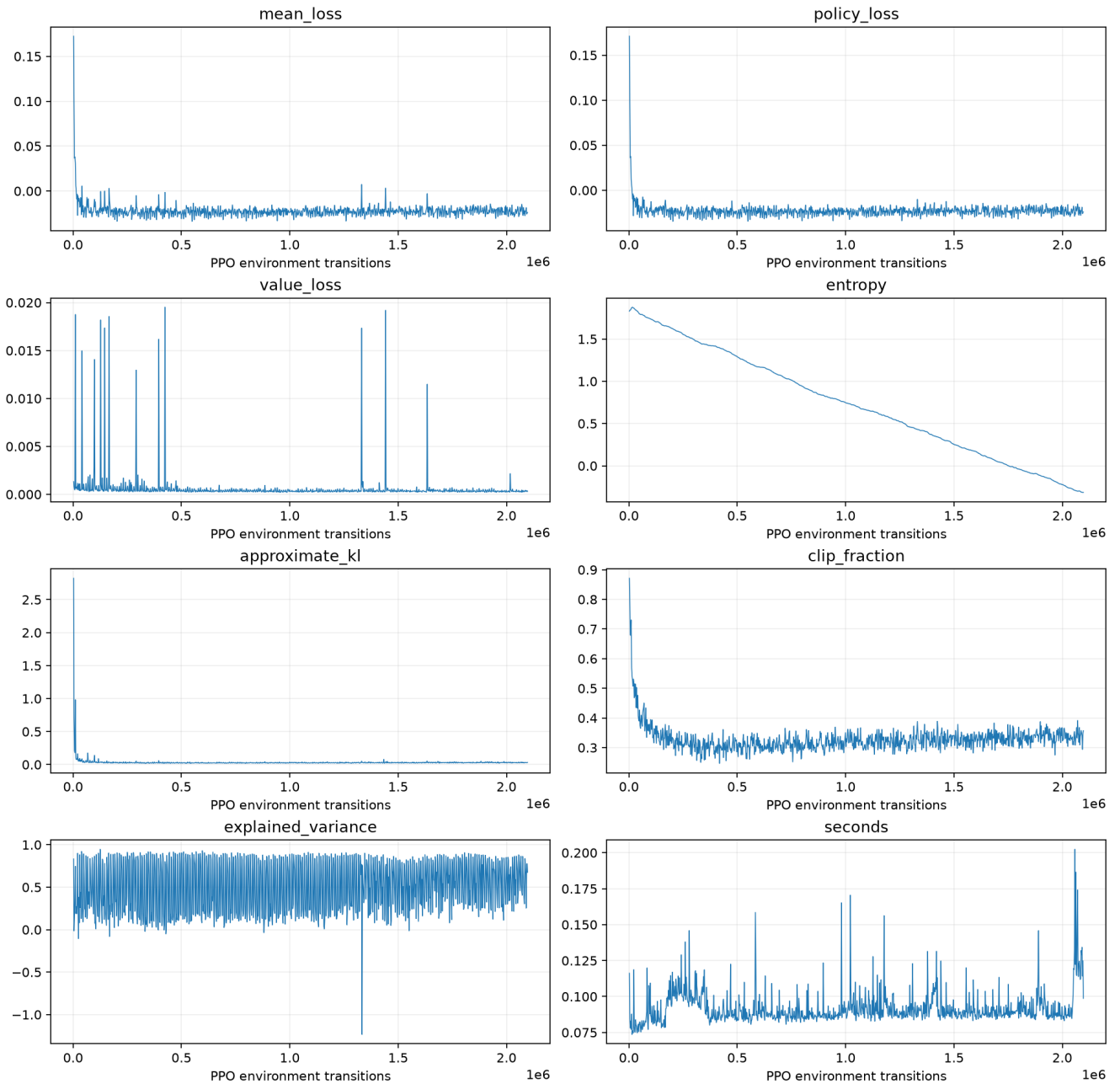
running: held-out seed physical measurements, PPO versus zero action



running: optimization evidence, not a substitute for skill acceptance



Actual PPO minibatch aggregates; full history including startup spikes



Source/artifact	Bytes	SHA-256
reward-learning.png	288,472	78aa85a12c5ccdaf6e9c987f82781ad8196d3a6174a2989059d5951896d6423a
ppo-losses.png	399,967	e50c299b8f0aae0ab329de5ce50f6d8af2176763919bb5752c5c46d06ed16ed8
browser-video.png	127,819	2bf8cdb2765722b3975b50fe52a12cda6e411f58be288bf787e6643a12d22033
studio-evaluation.png	83,283	1a9f7434f2b98ea3ebc035fd4e983db09ddc023e598d62877eedb0997acb5a08
comparison.mp4	893,717	522aa57a3cbdebb95650bbfd312cc491fca44753c0992f1fee50e90754dfbb18
api-video.mp4	967,405	cf1b3fc6abd932ae0693451993ff3fcf0f081a8046ab42c68e0f211966e23655

Reproduction commands

Run all stages from the repository root with the Studio server at <http://127.0.0.1:63317>. Replace YYYYYMMDD-HHMMSS consistently with a fresh timestamp; base and continuation runs must remain distinct.

This continuation resumes the Running v3 final checkpoint. Train the tracked base recipe first. The recorded base completed training but failed its skill gate, so the API runner exits nonzero for that skill failure. Retain its failed verdict. Reuse its completed checkpoint only if the API train operation succeeded and both checkpoint and sidecar exist. The guarded block accepts only the runner's explicit skill-failure outcome; missing evidence, failed training, and other runner errors stop the copy. Do not ignore failures unconditionally or resume after this block fails.

```
(
set -eu
test ! -e rlx/runs/studio/running/running-base-reproduction-YYYYMMDD-HHMMSS
test ! -e rlx/artifacts/scenarios-e2e-20260907/running-base-reproduction-YYYYMMDD-HHMMSS-api.json
test ! -e rlx/runs/studio/running/running-reproduction-YYYYMMDD-HHMMSS
base_status=0
node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
  running --recipe-json docs/remaining-scenarios-e2e/recipes/running-base.json --run running-base-
  reproduction-YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/running-base-
  reproduction-YYYYMMDD-HHMMSS-api.json --timeout-seconds 7200 || base_status=$?
python3 - rlx/artifacts/scenarios-e2e-20260907/running-base-reproduction-YYYYMMDD-HHMMSS-api.json running-
  base-reproduction-YYYYMMDD-HHMMSS "$base_status" <<'PY'

import json
import sys
from pathlib import Path
report = json.loads(Path(sys.argv[1]).read_text(encoding='utf-8'))
recipe = report.get('requestedRecipe') or {}
if recipe.get('experimentId') != 'running' or recipe.get('runName') != sys.argv[2]:
    raise SystemExit('Base API report does not match this run')
operations = {op['action']: op for op in report.get('operations', [])}
train = operations.get('train', {})
if (train.get('state') or {}).get('phase') != 'succeeded' or train.get('error'):
    raise SystemExit('Base training did not succeed; refusing checkpoint copy')
failure = report.get('failure')
if int(sys.argv[3]) != 0:
    expected = 'Full Running skill evaluation failed; render and export evidence were collected.'
    evaluation = (operations.get('eval', {}).get('state') or {}).get('evaluation') or {}
    if (failure or {}).get('message') != expected or evaluation.get('skill_status') != 'failed':
```

```

        raise SystemExit('Unexpected runner failure; refusing checkpoint copy')
    for action in ('render', 'export'):
        operation = operations.get(action, {})
        if (operation.get('state') or {}).get('phase') != 'succeeded' or operation.get('error'):
            raise SystemExit('Base evidence collection failed; refusing checkpoint copy')
    elif failure:
        raise SystemExit('Runner status contradicts API failure; refusing checkpoint copy')
PY
test -s rlx/runs/studio/running/running-base-reproduction-YYYYMMDD-HHMMSS/running.safetensors
test -s rlx/runs/studio/running/running-base-reproduction-YYYYMMDD-HHMMSS/running.safetensors.json
mkdir -p rlx/runs/studio/running/running-reproduction-YYYYMMDD-HHMMSS
cp rlx/runs/studio/running/running-base-reproduction-YYYYMMDD-HHMMSS/running.safetensors rlx/runs/studio/
    running/running-base-reproduction-YYYYMMDD-HHMMSS/running.safetensors.json rlx/runs/studio/running/
    running-reproduction-YYYYMMDD-HHMMSS/
)

```

After this block succeeds, the continuation runner below uses the target-local checkpoint and the selected recipe's reward weights. It does not reuse the base run's skill verdict.

Run from the repository root after starting the Studio server at <http://127.0.0.1:63317>:

```

node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
    running --recipe-json docs/remaining-scenarios-e2e/recipes/running.json --run running-reproduction-
    YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/running-reproduction-YYYYMMDD-HHMMSS-
    api.json --timeout-seconds 7200
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json docs/remaining-scenarios-e2e/
    recipes/running.json --run rlx/runs/studio/running/running-reproduction-YYYYMMDD-HHMMSS --output
    rlx/artifacts/scenarios-e2e-20260907/running-reproduction-YYYYMMDD-HHMMSS-audit --
    seeds 501 502 503 504 505 --render
node duck-viewer/scripts/verify-rlx-video.mjs --recipe-json docs/remaining-scenarios-e2e/recipes/
    running.json --run rlx/runs/studio/running/running-reproduction-YYYYMMDD-HHMMSS --output rlx/
    artifacts/scenarios-e2e-20260907/running-reproduction-YYYYMMDD-HHMMSS-audit --base-url http://
    127.0.0.1:63317 --playwright-package "${PLAYWRIGHT_PACKAGE:-/Users/ghu/community/package.json}"

```

The audit command exits nonzero when the strict skill gate fails. That nonzero exit is expected evidence for a failed attempt, not permission to rewrite the result as successful.

6. Stilt Walking: stilts-v3

Scenario status: PASS Complete API/audit/video evidence chain: verified

Recipe and effective audit environment

The exact requested recipe follows. These values describe training; they do not override the deterministic audit environment.

```

{
  "actionDelay": false,
  "checkpointInterval": 262144,
  "clipCoefficient": 0.2,
  "domainRand": false,
  "entropyCoefficient": 0,
  "evalSteps": 500,
  "experimentId": "stilts",

```

```

"gamma": 0.99,
"initialStd": 0.6,
"learningRate": 0.0001,
"locomotionForwardCommand": 0.25,
"maxEpisodeS": 10,
"maxGradNorm": 0.5,
"normalizeRewards": true,
"numEnvs": 16,
"numMinibatches": 4,
"numSteps": 128,
"obsNoise": false,
"profile": "full",
"randomYaw": false,
"renderSeconds": 10,
"resumeFromCheckpoint": true,
"rewardWeights": {
  "ang_vel_xy_penalty": 0.02,
  "feet_air_time": 4,
  "head_pose": 0.1,
  "track_ang_vel": 4,
  "track_lin_vel": 20,
  "upright": 1
},
"runName": "stilts-e2e-20260907-v3",
"seed": 7,
"stiltBlend": 0,
"stiltHeightCm": 2,
"stiltMassKg": 0.014,
"totalTimesteps": 1048576,
"updateEpochs": 4
}

```

Audit setting	Actual value
action_delay	no
actuator	xml
domain_rand	no
locomotion_forward_command	0.2500
max_episode_s	10
obs_noise	no
random_yaw	no
stilt_blend	0
stilt_height_cm	2
stilt_mass_kg	0.0140
swing_initial_angle_deg	0

Audit setting	Actual value
swing_initial_rate_rad_s	0
swing_planar_actions	no
weight_overrides	{'track_lin_vel': 20, 'track_ang_vel': 4, 'upright': 1, 'feet_air_time': 4, 'head_pose': 0.1, 'ang_vel_xy_penalty': 0.02}

The requested morphology is **2 cm** stilts, blend **0**, mass **0.0140 kg**, with a fixed forward command of **0.2500 m/s**.

Per-episode skill evidence

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
501	0	500	yes	1.0000	1.0000	0.2314	1.8273	0.9254	PASS
502	0	500	yes	1.0000	1.0000	0.2366	1.2806	0.9463	PASS
503	0	500	yes	1.0000	1.0000	0.2416	2.1831	0.9664	PASS
504	0	500	yes	1.0000	1.0000	0.2387	2.2887	0.9549	PASS
505	0	500	yes	1.0000	1.0000	0.2300	1.4662	0.9199	PASS

Per-episode physics details

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
501	0	0.6320 / 0.6420	0.3680 / 0.3580	0.0020	90	46 / 45	47 / 46
502	0	0.6360 / 0.6400	0.3640 / 0.3600	0.0020	92	47 / 47	48 / 48
503	0	0.6100 / 0.6440	0.3900 / 0.3560	0.0020	92	47 / 46	47 / 47
504	0	0.5900 / 0.6480	0.4100 / 0.3520	0.0020	90	47 / 45	47 / 46
505	0	0.6300 / 0.6500	0.3700 / 0.3500	0.0020	91	46 / 46	47 / 46

Per-episode failures

- **seed 501, episode 0:** none
- **seed 502, episode 0:** none
- **seed 503, episode 0:** none
- **seed 504, episode 0:** none
- **seed 505, episode 0:** none

Controls

Zero control

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
501	0	46	no	0.7609	1.0000	0.1575	0.1375	0.6300	FAIL
502	0	39	no	0.7436	1.0000	0.1781	0.1315	0.7122	FAIL
503	0	40	no	0.7500	1.0000	0.1750	0.1327	0.6999	FAIL
504	0	42	no	0.7619	1.0000	0.1663	0.1324	0.6652	FAIL
505	0	45	no	0.7556	1.0000	0.1666	0.1427	0.6663	FAIL

Physical details:

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
501	0	0.9348 / 0.9348	0.0652 / 0.0652	0.0652	0	2 / 2	2 / 2
502	0	0.9231 / 0.9231	0.0769 / 0.0769	0.0769	0	2 / 2	3 / 3
503	0	0.9250 / 0.9250	0.0750 / 0.0750	0.0750	0	2 / 2	2 / 2
504	0	0.9286 / 0.9286	0.0714 / 0.0714	0.0714	0	2 / 2	3 / 3
505	0	0.9778 / 0.9778	0.0222 / 0.0222	0.0222	0	0 / 0	1 / 1

Failures:

- **seed 501, episode 0:** command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; upright fraction below target
- **seed 502, episode 0:** command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; upright fraction below target
- **seed 503, episode 0:** command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; upright fraction below target
- **seed 504, episode 0:** command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; upright fraction below target
- **seed 505, episode 0:** both feet must lift off and touch down; command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

Random control

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
501	0	40	no	0.7750	1.0000	0.1722	0.1442	0.6886	FAIL
502	0	36	no	0.7500	1.0000	0.1662	0.1396	0.6647	FAIL
503	0	35	no	0.7429	1.0000	0.1939	0.1398	0.7757	FAIL
504	0	37	no	0.7568	1.0000	0.1840	0.1414	0.7360	FAIL
505	0	43	no	0.7907	1.0000	0.1695	0.1470	0.6778	FAIL

Physical details:

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
501	0	0.9750 / 0.9000	0.0250 / 0.1000	0.0250	0	0 / 2	1 / 2
502	0	0.9444 / 0.9167	0.0556 / 0.0833	0.0278	1	1 / 1	1 / 2
503	0	0.9429 / 0.9143	0.0571 / 0.0857	0.0571	0	1 / 2	1 / 2
504	0	0.9459 / 0.8919	0.0541 / 0.1081	0.0541	0	1 / 2	1 / 2
505	0	0.9535 / 0.9302	0.0465 / 0.0698	0.0233	2	1 / 2	2 / 2

Failures:

- **seed 501, episode 0:** both feet must lift off and touch down; command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 502, episode 0:** command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; upright fraction below target
- **seed 503, episode 0:** command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; upright fraction below target
- **seed 504, episode 0:** command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; upright fraction below target
- **seed 505, episode 0:** command-directed displacement below target; episode did not complete the required 10-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

Initial pretrained baseline

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
501	0	500	yes	1.0000	1.0000	0.2448	0.7163	0.9794	PASS

Physical details:

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
501	0	0.5980 / 0.6560	0.4020 / 0.3440	0.0020	100	51 / 50	51 / 51

Failures:

- **seed 501, episode 0:** none

Physics metric means

Control	Seed	Metric	Mean
trained	501	command_forward_m_s	0.2500
trained	501	command_lateral_m_s	0.0000
trained	501	command_yaw_rad_s	0.0000
trained	501	forward_speed_m_s	0.2314
trained	501	heading_forward_speed_m_s	0.2314
trained	501	heading_forward_x	0.8049
trained	501	heading_forward_y	0.4870
trained	501	heading_lateral_speed_m_s	0.0038
trained	501	left_foot_contact	0.6320
trained	501	right_foot_contact	0.6420
trained	501	stilt_blend	0.0000
trained	501	stilt_height_cm	2.0000
trained	501	stilt_mass_kg	0.0140
trained	501	upright	0.9993
trained	501	world_x_m	0.9699
trained	501	world_y_m	0.4088
trained	502	command_forward_m_s	0.2500
trained	502	command_lateral_m_s	0.0000
trained	502	command_yaw_rad_s	0.0000
trained	502	forward_speed_m_s	0.2366

Control	Seed	Metric	Mean
trained	502	heading_forward_speed_m_s	0.2366
trained	502	heading_forward_x	0.5459
trained	502	heading_forward_y	0.6321
trained	502	heading_lateral_speed_m_s	-0.0040
trained	502	left_foot_contact	0.6360
trained	502	right_foot_contact	0.6400
trained	502	stilt_blend	0.0000
trained	502	stilt_height_cm	2.0000
trained	502	stilt_mass_kg	0.0140
trained	502	upright	0.9989
trained	502	world_x_m	0.9195
trained	502	world_y_m	0.5335
trained	503	command_forward_m_s	0.2500
trained	503	command_lateral_m_s	0.0000
trained	503	command_yaw_rad_s	0.0000
trained	503	forward_speed_m_s	0.2416
trained	503	heading_forward_speed_m_s	0.2416
trained	503	heading_forward_x	0.8902
trained	503	heading_forward_y	0.2525
trained	503	heading_lateral_speed_m_s	-0.0047
trained	503	left_foot_contact	0.6100
trained	503	right_foot_contact	0.6440
trained	503	stilt_blend	0.0000
trained	503	stilt_height_cm	2.0000
trained	503	stilt_mass_kg	0.0140
trained	503	upright	0.9992
trained	503	world_x_m	1.1818
trained	503	world_y_m	0.0529
trained	504	command_forward_m_s	0.2500
trained	504	command_lateral_m_s	0.0000
trained	504	command_yaw_rad_s	0.0000
trained	504	forward_speed_m_s	0.2387
trained	504	heading_forward_speed_m_s	0.2387

Control	Seed	Metric	Mean
trained	504	heading_forward_x	0.9558
trained	504	heading_forward_y	0.2127
trained	504	heading_lateral_speed_m_s	-0.0051
trained	504	left_foot_contact	0.5900
trained	504	right_foot_contact	0.6480
trained	504	stilt_blend	0.0000
trained	504	stilt_height_cm	2.0000
trained	504	stilt_mass_kg	0.0140
trained	504	upright	0.9992
trained	504	world_x_m	1.1402
trained	504	world_y_m	0.1147
trained	505	command_forward_m_s	0.2500
trained	505	command_lateral_m_s	0.0000
trained	505	command_yaw_rad_s	0.0000
trained	505	forward_speed_m_s	0.2300
trained	505	heading_forward_speed_m_s	0.2300
trained	505	heading_forward_x	0.6269
trained	505	heading_forward_y	0.5010
trained	505	heading_lateral_speed_m_s	-0.0032
trained	505	left_foot_contact	0.6300
trained	505	right_foot_contact	0.6500
trained	505	stilt_blend	0.0000
trained	505	stilt_height_cm	2.0000
trained	505	stilt_mass_kg	0.0140
trained	505	upright	0.9990
trained	505	world_x_m	0.9842
trained	505	world_y_m	0.3125
zero	501	command_forward_m_s	0.2500
zero	501	command_lateral_m_s	0.0000
zero	501	command_yaw_rad_s	0.0000
zero	501	forward_speed_m_s	0.1575
zero	501	heading_forward_speed_m_s	0.1575
zero	501	heading_forward_x	1.0000

Control	Seed	Metric	Mean
zero	501	heading_forward_y	0.0065
zero	501	heading_lateral_speed_m_s	0.0011
zero	501	left_foot_contact	0.9348
zero	501	right_foot_contact	0.9348
zero	501	stilt_blend	0.0000
zero	501	stilt_height_cm	2.0000
zero	501	stilt_mass_kg	0.0140
zero	501	upright	0.9109
zero	501	world_x_m	0.0276
zero	501	world_y_m	0.0003
zero	502	command_forward_m_s	0.2500
zero	502	command_lateral_m_s	0.0000
zero	502	command_yaw_rad_s	0.0000
zero	502	forward_speed_m_s	0.1781
zero	502	heading_forward_speed_m_s	0.1781
zero	502	heading_forward_x	1.0000
zero	502	heading_forward_y	0.0087
zero	502	heading_lateral_speed_m_s	0.0006
zero	502	left_foot_contact	0.9231
zero	502	right_foot_contact	0.9231
zero	502	stilt_blend	0.0000
zero	502	stilt_height_cm	2.0000
zero	502	stilt_mass_kg	0.0140
zero	502	upright	0.9001
zero	502	world_x_m	0.0283
zero	502	world_y_m	-0.0008
zero	503	command_forward_m_s	0.2500
zero	503	command_lateral_m_s	0.0000
zero	503	command_yaw_rad_s	0.0000
zero	503	forward_speed_m_s	0.1750
zero	503	heading_forward_speed_m_s	0.1750
zero	503	heading_forward_x	1.0000
zero	503	heading_forward_y	-0.0055

Control	Seed	Metric	Mean
zero	503	heading_lateral_speed_m_s	-0.0012
zero	503	left_foot_contact	0.9250
zero	503	right_foot_contact	0.9250
zero	503	stilt_blend	0.0000
zero	503	stilt_height_cm	2.0000
zero	503	stilt_mass_kg	0.0140
zero	503	upright	0.9002
zero	503	world_x_m	0.0274
zero	503	world_y_m	-0.0014
zero	504	command_forward_m_s	0.2500
zero	504	command_lateral_m_s	0.0000
zero	504	command_yaw_rad_s	0.0000
zero	504	forward_speed_m_s	0.1663
zero	504	heading_forward_speed_m_s	0.1663
zero	504	heading_forward_x	0.9999
zero	504	heading_forward_y	0.0120
zero	504	heading_lateral_speed_m_s	7.341e-05
zero	504	left_foot_contact	0.9286
zero	504	right_foot_contact	0.9286
zero	504	stilt_blend	0.0000
zero	504	stilt_height_cm	2.0000
zero	504	stilt_mass_kg	0.0140
zero	504	upright	0.9105
zero	504	world_x_m	0.0275
zero	504	world_y_m	-0.0003
zero	505	command_forward_m_s	0.2500
zero	505	command_lateral_m_s	0.0000
zero	505	command_yaw_rad_s	0.0000
zero	505	forward_speed_m_s	0.1666
zero	505	heading_forward_speed_m_s	0.1666
zero	505	heading_forward_x	0.9994
zero	505	heading_forward_y	-0.0307
zero	505	heading_lateral_speed_m_s	0.0032

Control	Seed	Metric	Mean
zero	505	left_foot_contact	0.9778
zero	505	right_foot_contact	0.9778
zero	505	stilt_blend	0.0000
zero	505	stilt_height_cm	2.0000
zero	505	stilt_mass_kg	0.0140
zero	505	upright	0.9051
zero	505	world_x_m	0.0305
zero	505	world_y_m	0.0004
initial	501	command_forward_m_s	0.2500
initial	501	command_lateral_m_s	0.0000
initial	501	command_yaw_rad_s	0.0000
initial	501	forward_speed_m_s	0.2448
initial	501	heading_forward_speed_m_s	0.2448
initial	501	heading_forward_x	0.3222
initial	501	heading_forward_y	-0.3258
initial	501	heading_lateral_speed_m_s	0.0010
initial	501	left_foot_contact	0.5980
initial	501	right_foot_contact	0.6560
initial	501	stilt_blend	0.0000
initial	501	stilt_height_cm	2.0000
initial	501	stilt_mass_kg	0.0140
initial	501	upright	0.9977
initial	501	world_x_m	0.8582
initial	501	world_y_m	-0.3439

PPO transitions and losses

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	512
Optimizer minibatches	8,192
Observed transitions	1,048,576
Requested transitions	1,048,576
Monotonic update steps	yes

Training evidence	Value
Training evidence gate	PASS
Maximum approximate KL	0.7690
Last-100 mean_loss	-0.0193
Last-100 policy_loss	-0.0206
Last-100 value_loss	0.0013
Last-100 entropy	1.1074
Last-100 approximate_kl	0.0191
Last-100 clip_fraction	0.2322
Last-100 explained_variance	0.1992

Transitions	Steps	Raw return	Skill	Failures
0	500	10,431.73	PASS	none
262,144	500	10,886.09	PASS	none
524,288	500	10,934.60	PASS	none
786,432	500	11,016.86	PASS	none
1,048,576	500	11,189.72	PASS	none

Initialization and PPO contribution

Initialization evidence	Value
Source checkpoint	/Volumes/ExternalSSD/geoagent/microduck-lab/rlx/runs/studio/stilts/stilts-e2e-20260907-v3/stilts.safetensors
Source SHA-256	0f0c8e2291ee58018d6c60de3107330da4be6d6a50dedc40b49752afc5edf4bc
Teacher-assisted	n/a
Initial policy role	pretrained baseline, not a null control
Comparison seed	501
PPO raw-return change vs initialization	757.9966

This run continues a pretrained policy. Its initial skill is not credited to PPO. A nonzero parameter change proves optimization occurred, not that it improved the initializer. Compare the baseline and final physical outcomes and raw returns before attributing any benefit to PPO.

ONNX, controls, and source identity

Check	Value
Policy SHA-256	4e1bdf737c10ef253cc209097509bf3229c08496e71e455b15d2683f37e6cb24
ONNX input	batch x 61

Check	Value
ONNX output	batch x 14
ONNX contract	PASS
Native/ONNX max absolute error	9.537e-07
Parameter L2 change	6.6147
Negative controls failed as required	yes
Audit scope	Nominal local simulation only; no hardware certification. No assisted Swing starts.

API workflow

Action	Accepted	Phase	Exit code	Finished
train	yes	succeeded	0	2026-09-08T01:23:38.059Z
eval	yes	succeeded	0	2026-09-08T01:23:48.108Z
render	yes	succeeded	0	2026-09-08T01:24:00.063Z
export	yes	succeeded	0	2026-09-08T01:24:05.307Z

Video and graph evidence

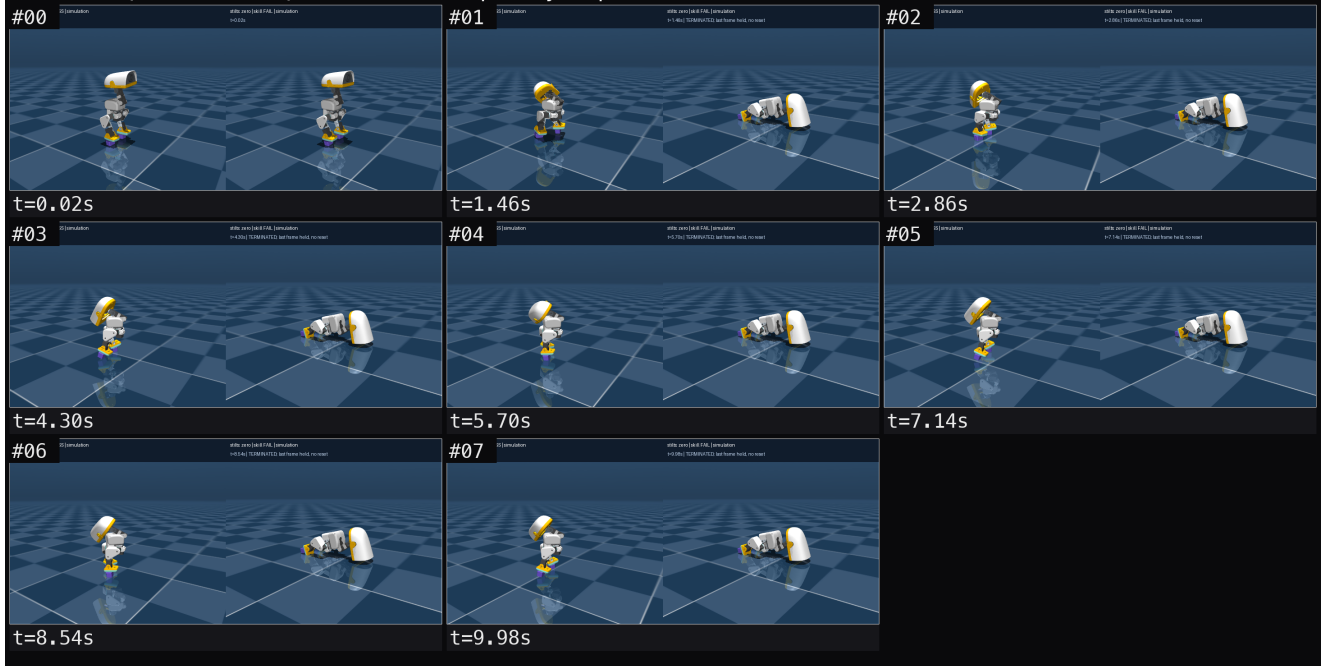
Video validation	Value
Validation status	PASS
API bytes equal local render	yes
HTTP byte range	yes
Browser playback	{"currentTime": 5, "duration": 10, "error": null, "height": 360, "width": 640}

File	Codec	Dimensions	FPS	Frames	Seconds	SHA-256
api-video.mp4	h264	640 x 360	25/1	250	10.000000	7cfddb35396c563d568d805dcda302aa83c41b805d2f3deaaaae89d85a67ada8
comparison.mp4	h264	960 x 412	25/1	250	10.000000	ae6183848a1960f080a8929675406e3681fd35e090c76e7e7240a8b3ea41d512

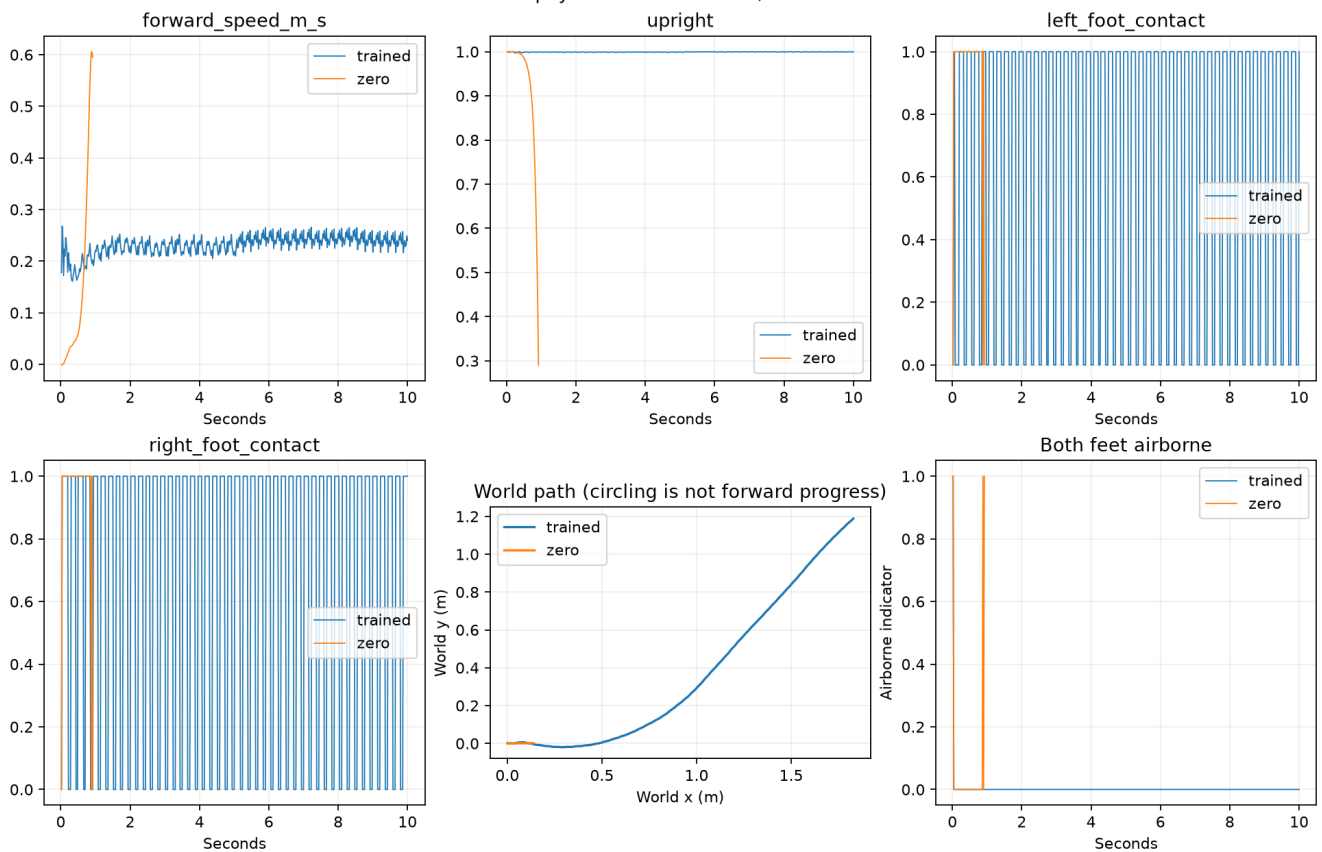
- [comparison_sheet.png](#)
- [physical-tracking.png](#)
- [reward-learning.png](#)
- [ppo-losses.png](#)
- [browser-video.png](#)
- [studio-evaluation.png](#)
- [comparison.mp4](#)
- [api-video.mp4](#)
- [audit.json](#)
- [video-validation.json](#)

- stilts-v3-api.json
- stilts-v3-api.log

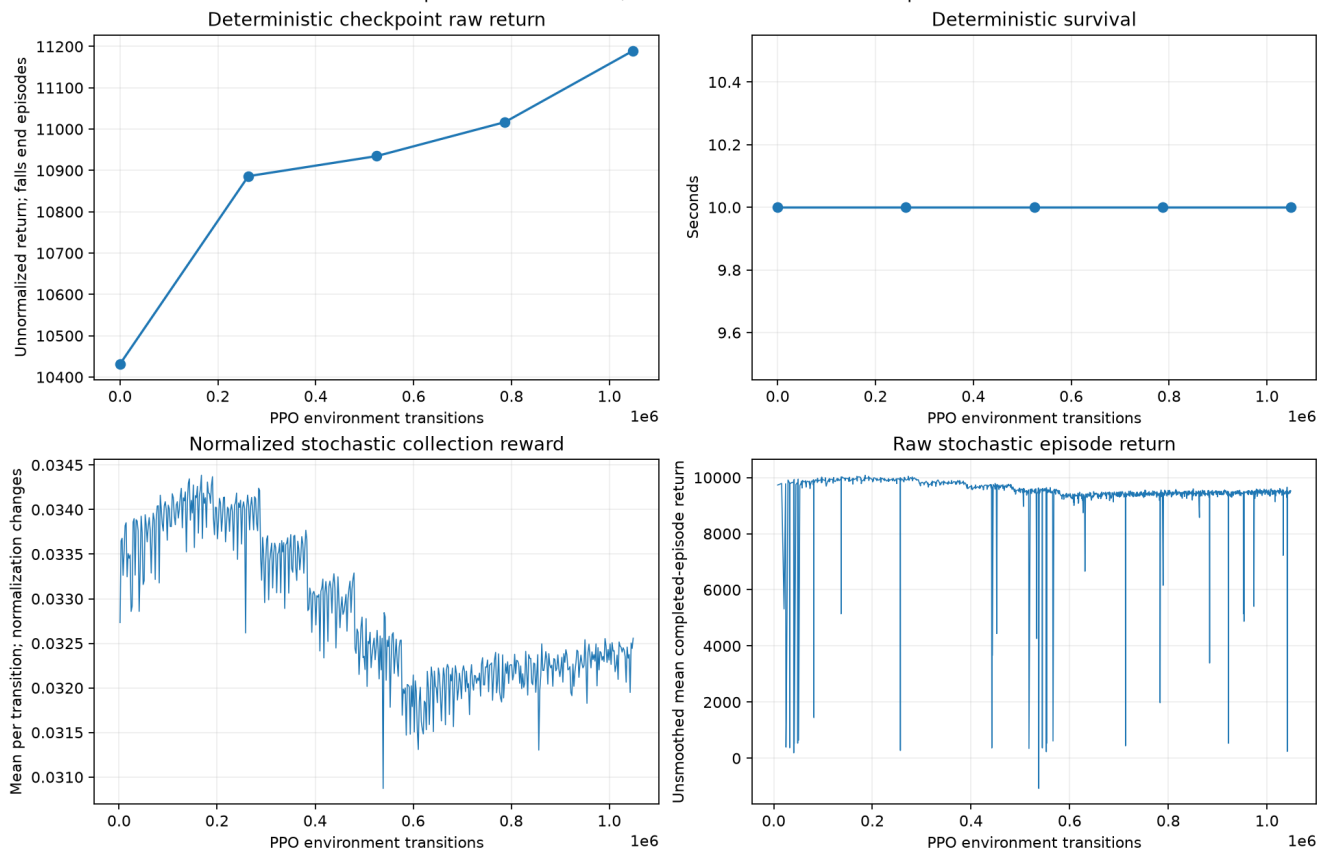
stilts: trained ONNX versus zero action
No resets, assistance, or kinematic policy replacement.



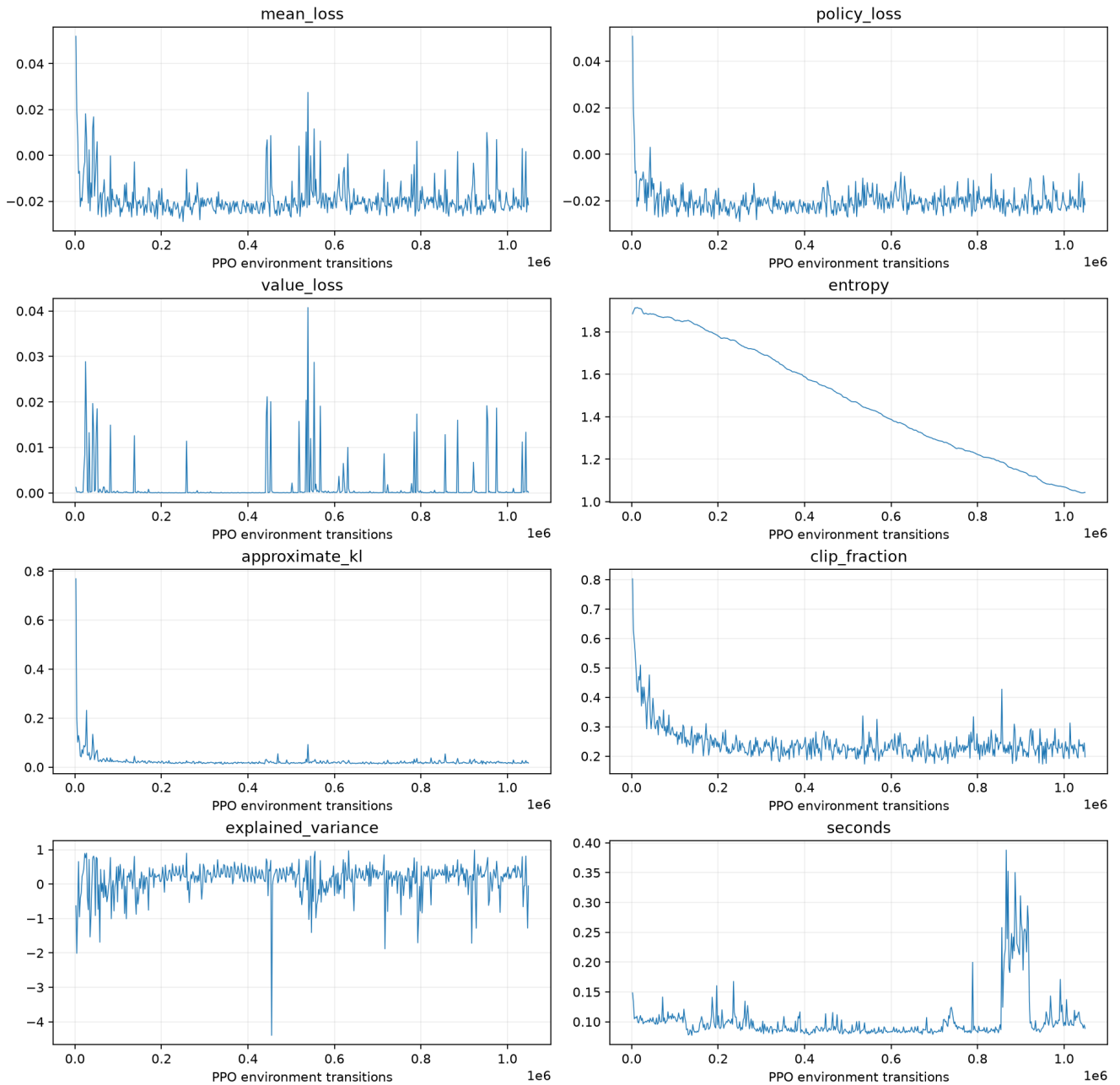
stilts: held-out seed physical measurements, PPO versus zero action



stilts: optimization evidence, not a substitute for skill acceptance



Actual PPO minibatch aggregates; full history including startup spikes



Source/artifact	Bytes	SHA-256
physical-tracking.png	341,339	13130001961c57f258af22722954403f966b17140238f7b3a6a5699d812b5bd1
reward-learning.png	253,527	d09c4c1592d0cc76f32f37401a377351604083ffaaead2002819a07312405893
ppo-losses.png	421,287	52cb03b00487e24c5aef90112083973782c7fd5c4216698a6f2ce83940e00e97
browser-video.png	122,025	7355147cd34c189fc665bf507ed9f6de10b96d207318c71c41d2a4a7e3f2a66c
studio-evaluation.png	82,894	fbda1e3860662d67cd7531af14e697d22b750c278fddc34f463c1a6d785338f4
comparison.mp4	474,477	ae6183848a1960f080a8929675406e3681fd35e090c76e7e7240a8b3ea41d512
api-video.mp4	442,905	7cfddb35396c563d568d805dcda302aa83c41b805d2f3deaaaae89d85a67ada8

Reproduction commands

Run all stages from the repository root with the Studio server at <http://127.0.0.1:63317>. Replace YYYMMDD-HHMMSS consistently with a fresh timestamp; base and continuation runs must remain distinct.

This continuation resumes the Stilt v2 final checkpoint. Train the tracked base recipe first. The recorded base completed training but failed its skill gate, so the API runner exits nonzero for that skill failure. Retain its failed verdict. Reuse its completed checkpoint only if the API train operation succeeded and both checkpoint and sidecar exist. The guarded block accepts only the runner's explicit skill-failure outcome; missing evidence, failed training, and other runner errors stop the copy. Do not ignore failures unconditionally or resume after this block fails.

```
(
set -eu
test ! -e rlx/runs/studio/stilts/stilts-base-reproduction-YYYMMDD-HHMMSS
test ! -e rlx/artifacts/scenarios-e2e-20260907/stilts-base-reproduction-YYYMMDD-HHMMSS-api.json
test ! -e rlx/runs/studio/stilts/stilts-reproduction-YYYMMDD-HHMMSS
base_status=0
node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
  stilts --recipe-json docs/remaining-scenarios-e2e/recipes/stilts-base.json --run stilts-base-
  reproduction-YYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/stilts-base-
  reproduction-YYYMMDD-HHMMSS-api.json --timeout-seconds 7200 || base_status=$?
python3 - rlx/artifacts/scenarios-e2e-20260907/stilts-base-reproduction-YYYMMDD-HHMMSS-api.json stilts-
  base-reproduction-YYYMMDD-HHMMSS "$base_status" <<'PY'

import json
import sys
from pathlib import Path
report = json.loads(Path(sys.argv[1]).read_text(encoding='utf-8'))
recipe = report.get('requestedRecipe') or {}
if recipe.get('experimentId') != 'stilts' or recipe.get('runName') != sys.argv[2]:
    raise SystemExit('Base API report does not match this run')
operations = {op['action']: op for op in report.get('operations', [])}
train = operations.get('train', {})
if (train.get('state') or {}).get('phase') != 'succeeded' or train.get('error'):
    raise SystemExit('Base training did not succeed; refusing checkpoint copy')
failure = report.get('failure')
if int(sys.argv[3]) != 0:
    expected = 'Full Stilts skill evaluation failed; render and export evidence were collected.'
    evaluation = (operations.get('eval', {}).get('state') or {}).get('evaluation') or {}
```

```

    if (failure or {}).get('message') != expected or evaluation.get('skill_status') != 'failed':
        raise SystemExit('Unexpected runner failure; refusing checkpoint copy')
    for action in ('render', 'export'):
        operation = operations.get(action, {})
        if (operation.get('state') or {}).get('phase') != 'succeeded' or operation.get('error'):
            raise SystemExit('Base evidence collection failed; refusing checkpoint copy')
elif failure:
    raise SystemExit('Runner status contradicts API failure; refusing checkpoint copy')
PY
test -s rlx/runs/studio/stilts/stilts-base-reproduction-YYYYMMDD-HHMMSS/stilts.safetensors
test -s rlx/runs/studio/stilts/stilts-base-reproduction-YYYYMMDD-HHMMSS/stilts.safetensors.json
mkdir -p rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS
cp rlx/runs/studio/stilts/stilts-base-reproduction-YYYYMMDD-HHMMSS/stilts.safetensors rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS/stilts.safetensors.json rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS/
)

```

After this block succeeds, the continuation runner below uses the target-local checkpoint and the selected recipe's reward weights. It does not reuse the base run's skill verdict.

Run from the repository root after starting the Studio server at <http://127.0.0.1:63317>:

```

node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
stilts --recipe-json docs/remaining-scenarios-e2e/recipes/stilts.json --run stilts-reproduction-
YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-
api.json --timeout-seconds 7200
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json docs/remaining-scenarios-e2e/
recipes/stilts.json --run rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS --output rlx/
artifacts/scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-audit --seeds 501 502 503 504
505 --render
node duck-viewer/scripts/verify-rlx-video.mjs --recipe-json docs/remaining-scenarios-e2e/recipes/
stilts.json --run rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS --output rlx/
artifacts/scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-audit --base-url http://
127.0.0.1:63317 --playwright-package "${PLAYWRIGHT_PACKAGE:-/Users/ghu/community/package.json}"

```

The audit command exits nonzero when the strict skill gate fails. That nonzero exit is expected evidence for a failed attempt, not permission to rewrite the result as successful.

7. Swing: swing-v3

Scenario status: PASS Complete API/audit/video evidence chain: verified

Recipe and effective audit environment

The exact requested recipe follows. These values describe training; they do not override the deterministic audit environment.

```

{
  "actionDelay": false,
  "checkpointInterval": 262144,
  "clipCoefficient": 0.05,
  "domainRand": false,
  "entropyCoefficient": 0,
  "evalSteps": 1200,

```

```

"experimentId": "swing",
"freezeObservationNormalization": true,
"gamma": 0.995,
"initialStd": 0.1,
"learningRate": 3e-06,
"maxEpisodeS": 24,
"maxGradNorm": 0.5,
"normalizeRewards": true,
"numEnvs": 16,
"numMinibatches": 4,
"numSteps": 256,
"obsNoise": false,
"profile": "full",
"randomYaw": false,
"renderSeconds": 24,
"resumeFromCheckpoint": true,
"runName": "swing-e2e-20260907-v3",
"seed": 7,
"swingInitialAngleDeg": 0,
"swingInitialRateRadS": 0,
"swingMinSpanDeg": 150,
"swingPlanarActions": true,
"totalTimesteps": 524288,
"updateEpochs": 2
}

```

Audit setting	Actual value
action_delay	no
actuator	xml
domain_rand	no
locomotion_forward_command	n/a
max_episode_s	24
obs_noise	no
random_yaw	no
stilt_blend	0
stilt_height_cm	2
stilt_mass_kg	n/a
swing_initial_angle_deg	0
swing_initial_rate_rad_s	0
swing_planar_actions	yes
weight_overrides	{}

Training assistance is explicit in the recipe: initial angle **0 degrees**, initial rate **0 rad/s**, planar actions **yes**. The independent audit must start at rest (0 degrees, 0 rad/s); assisted training starts never count as acceptance evidence.

Per-episode skill evidence

Seed	Episode	Steps	Complete	Bidirectional span	Negative peak	Positive peak	Tensioned	Valid geometry	Verdict
501	0	1,200	yes	162.3216	-81.1608	82.9435	1.0000	1.0000	PASS
502	0	1,200	yes	162.3216	-81.1608	82.9435	1.0000	1.0000	PASS
503	0	1,200	yes	162.3216	-81.1608	82.9435	1.0000	1.0000	PASS
504	0	1,200	yes	162.3216	-81.1608	82.9435	1.0000	1.0000	PASS
505	0	1,200	yes	162.3216	-81.1608	82.9435	1.0000	1.0000	PASS

Per-episode physics details

Seed	Episode	Max lateral	Max alignment	Min/max string length	Min spring tension
501	0	0.0090	0.0461	0.3801 / 0.3867	0.2090
502	0	0.0090	0.0461	0.3801 / 0.3867	0.2090
503	0	0.0090	0.0461	0.3801 / 0.3867	0.2090
504	0	0.0090	0.0461	0.3801 / 0.3867	0.2090
505	0	0.0090	0.0461	0.3801 / 0.3867	0.2090

Per-episode failures

- **seed 501, episode 0:** none
- **seed 502, episode 0:** none
- **seed 503, episode 0:** none
- **seed 504, episode 0:** none
- **seed 505, episode 0:** none

Controls

Zero control

Seed	Episode	Steps	Complete	Bidirectional span	Negative peak	Positive peak	Tensioned	Valid geometry	Verdict
501	0	1,200	yes	0.4957	-0.2478	0.2667	1.0000	1.0000	FAIL
502	0	1,200	yes	0.4957	-0.2478	0.2667	1.0000	1.0000	FAIL
503	0	1,200	yes	0.4957	-0.2478	0.2667	1.0000	1.0000	FAIL
504	0	1,200	yes	0.4957	-0.2478	0.2667	1.0000	1.0000	FAIL
505	0	1,200	yes	0.4957	-0.2478	0.2667	1.0000	1.0000	FAIL

Physical details:

Seed	Episode	Max lateral	Max alignment	Min/max string length	Min spring tension
501	0	5.015e-07	6.736e-10	0.3821 / 0.3822	4.2705
502	0	5.015e-07	6.736e-10	0.3821 / 0.3822	4.2705
503	0	5.015e-07	6.736e-10	0.3821 / 0.3822	4.2705
504	0	5.015e-07	6.736e-10	0.3821 / 0.3822	4.2705
505	0	5.015e-07	6.736e-10	0.3821 / 0.3822	4.2705

Failures:

- **seed 501, episode 0:** bidirectional span below target
- **seed 502, episode 0:** bidirectional span below target
- **seed 503, episode 0:** bidirectional span below target
- **seed 504, episode 0:** bidirectional span below target
- **seed 505, episode 0:** bidirectional span below target

Random control

Seed	Episode	Steps	Complete	Bidirectional span	Negative peak	Positive peak	Tensioned	Valid geometry	Verdict
501	0	1,200	yes	0.8067	-0.4393	0.4034	1.0000	1.0000	FAIL
502	0	1,200	yes	0.8067	-0.4393	0.4034	1.0000	1.0000	FAIL
503	0	1,200	yes	0.8067	-0.4393	0.4034	1.0000	1.0000	FAIL
504	0	1,200	yes	0.8067	-0.4393	0.4034	1.0000	1.0000	FAIL
505	0	1,200	yes	0.8067	-0.4393	0.4034	1.0000	1.0000	FAIL

Physical details:

Seed	Episode	Max lateral	Max alignment	Min/max string length	Min spring tension
501	0	1.001e-06	3.282e-08	0.3821 / 0.3823	4.2690
502	0	1.001e-06	3.282e-08	0.3821 / 0.3823	4.2690
503	0	1.001e-06	3.282e-08	0.3821 / 0.3823	4.2690
504	0	1.001e-06	3.282e-08	0.3821 / 0.3823	4.2690
505	0	1.001e-06	3.282e-08	0.3821 / 0.3823	4.2690

Failures:

- **seed 501, episode 0:** bidirectional span below target
- **seed 502, episode 0:** bidirectional span below target
- **seed 503, episode 0:** bidirectional span below target
- **seed 504, episode 0:** bidirectional span below target
- **seed 505, episode 0:** bidirectional span below target

Initial pretrained baseline

Seed	Episode	Steps	Complete	Bidirectional span	Negative peak	Positive peak	Tensioned	Valid geometry	Verdict
501	0	1,200	yes	154.6207	-77.3103	78.8132	1.0000	1.0000	PASS

Physical details:

Seed	Episode	Max lateral	Max alignment	Min/max string length	Min spring tension
501	0	0.0087	0.0061	0.3802 / 0.3867	0.3058

Failures:

- seed 501, episode 0: none

Physics metric means

Control	Seed	Metric	Mean
trained	501	alignment_penalty	0.0010
trained	501	lateral_offset_m	5.340e-05
trained	501	string_imbalance_m	3.895e-05
trained	501	string_left_m	0.3825
trained	501	string_left_tension_n	4.9731
trained	501	string_right_m	0.3825
trained	501	string_right_tension_n	4.9823
trained	501	string_slack_m	0.0000
trained	501	swing_abs_angle_deg	26.1614
trained	501	swing_angle_deg	-0.8205
trained	501	swing_rate_rad_s	-0.0078
trained	501	valid_geometry	1.0000
trained	502	alignment_penalty	0.0010
trained	502	lateral_offset_m	5.340e-05
trained	502	string_imbalance_m	3.895e-05
trained	502	string_left_m	0.3825
trained	502	string_left_tension_n	4.9731
trained	502	string_right_m	0.3825
trained	502	string_right_tension_n	4.9823
trained	502	string_slack_m	0.0000
trained	502	swing_abs_angle_deg	26.1614

Control	Seed	Metric	Mean
trained	502	swing_angle_deg	-0.8205
trained	502	swing_rate_rad_s	-0.0078
trained	502	valid_geometry	1.0000
trained	503	alignment_penalty	0.0010
trained	503	lateral_offset_m	5.340e-05
trained	503	string_imbalance_m	3.895e-05
trained	503	string_left_m	0.3825
trained	503	string_left_tension_n	4.9731
trained	503	string_right_m	0.3825
trained	503	string_right_tension_n	4.9823
trained	503	string_slack_m	0.0000
trained	503	swing_abs_angle_deg	26.1614
trained	503	swing_angle_deg	-0.8205
trained	503	swing_rate_rad_s	-0.0078
trained	503	valid_geometry	1.0000
trained	504	alignment_penalty	0.0010
trained	504	lateral_offset_m	5.340e-05
trained	504	string_imbalance_m	3.895e-05
trained	504	string_left_m	0.3825
trained	504	string_left_tension_n	4.9731
trained	504	string_right_m	0.3825
trained	504	string_right_tension_n	4.9823
trained	504	string_slack_m	0.0000
trained	504	swing_abs_angle_deg	26.1614
trained	504	swing_angle_deg	-0.8205
trained	504	swing_rate_rad_s	-0.0078
trained	504	valid_geometry	1.0000
trained	505	alignment_penalty	0.0010
trained	505	lateral_offset_m	5.340e-05
trained	505	string_imbalance_m	3.895e-05
trained	505	string_left_m	0.3825
trained	505	string_left_tension_n	4.9731
trained	505	string_right_m	0.3825

Control	Seed	Metric	Mean
trained	505	string_right_tension_n	4.9823
trained	505	string_slack_m	0.0000
trained	505	swing_abs_angle_deg	26.1614
trained	505	swing_angle_deg	-0.8205
trained	505	swing_rate_rad_s	-0.0078
trained	505	valid_geometry	1.0000
zero	501	alignment_penalty	1.797e-10
zero	501	lateral_offset_m	-2.003e-09
zero	501	string_imbalance_m	3.570e-07
zero	501	string_left_m	0.3822
zero	501	string_left_tension_n	4.3908
zero	501	string_right_m	0.3822
zero	501	string_right_tension_n	4.3915
zero	501	string_slack_m	0.0000
zero	501	swing_abs_angle_deg	0.1228
zero	501	swing_angle_deg	-0.0017
zero	501	swing_rate_rad_s	9.188e-06
zero	501	valid_geometry	1.0000
zero	502	alignment_penalty	1.797e-10
zero	502	lateral_offset_m	-2.003e-09
zero	502	string_imbalance_m	3.570e-07
zero	502	string_left_m	0.3822
zero	502	string_left_tension_n	4.3908
zero	502	string_right_m	0.3822
zero	502	string_right_tension_n	4.3915
zero	502	string_slack_m	0.0000
zero	502	swing_abs_angle_deg	0.1228
zero	502	swing_angle_deg	-0.0017
zero	502	swing_rate_rad_s	9.188e-06
zero	502	valid_geometry	1.0000
zero	503	alignment_penalty	1.797e-10
zero	503	lateral_offset_m	-2.003e-09
zero	503	string_imbalance_m	3.570e-07

Control	Seed	Metric	Mean
zero	503	string_left_m	0.3822
zero	503	string_left_tension_n	4.3908
zero	503	string_right_m	0.3822
zero	503	string_right_tension_n	4.3915
zero	503	string_slack_m	0.0000
zero	503	swing_abs_angle_deg	0.1228
zero	503	swing_angle_deg	-0.0017
zero	503	swing_rate_rad_s	9.188e-06
zero	503	valid_geometry	1.0000
zero	504	alignment_penalty	1.797e-10
zero	504	lateral_offset_m	-2.003e-09
zero	504	string_imbalance_m	3.570e-07
zero	504	string_left_m	0.3822
zero	504	string_left_tension_n	4.3908
zero	504	string_right_m	0.3822
zero	504	string_right_tension_n	4.3915
zero	504	string_slack_m	0.0000
zero	504	swing_abs_angle_deg	0.1228
zero	504	swing_angle_deg	-0.0017
zero	504	swing_rate_rad_s	9.188e-06
zero	504	valid_geometry	1.0000
zero	505	alignment_penalty	1.797e-10
zero	505	lateral_offset_m	-2.003e-09
zero	505	string_imbalance_m	3.570e-07
zero	505	string_left_m	0.3822
zero	505	string_left_tension_n	4.3908
zero	505	string_right_m	0.3822
zero	505	string_right_tension_n	4.3915
zero	505	string_slack_m	0.0000
zero	505	swing_abs_angle_deg	0.1228
zero	505	swing_angle_deg	-0.0017
zero	505	swing_rate_rad_s	9.188e-06
zero	505	valid_geometry	1.0000

Control	Seed	Metric	Mean
initial	501	alignment_penalty	0.0004
initial	501	lateral_offset_m	-1.971e-05
initial	501	string_imbalance_m	2.757e-05
initial	501	string_left_m	0.3825
initial	501	string_left_tension_n	4.9390
initial	501	string_right_m	0.3825
initial	501	string_right_tension_n	4.9425
initial	501	string_slack_m	0.0000
initial	501	swing_abs_angle_deg	23.4740
initial	501	swing_angle_deg	-0.4467
initial	501	swing_rate_rad_s	0.0451
initial	501	valid_geometry	1.0000

PPO transitions and losses

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	128
Optimizer minibatches	1,024
Observed transitions	524,288
Requested transitions	524,288
Monotonic update steps	yes
Training evidence gate	PASS
Maximum approximate KL	0.0080
Last-100 mean_loss	0.3728
Last-100 policy_loss	-0.0003
Last-100 value_loss	0.3731
Last-100 entropy	-12.3715
Last-100 approximate_kl	0.0005
Last-100 clip_fraction	0.0969
Last-100 explained_variance	-0.8999

Transitions	Steps	Raw return	Skill	Failures
0	1,200	6,126.39	PASS	none

Transitions	Steps	Raw return	Skill	Failures
262,144	1,200	5,797.53	FAIL	invalid swing geometry
524,288	1,200	7,016.89	PASS	none

Initialization and PPO contribution

Initialization evidence	Value
Source checkpoint	/Volumes/ExternalSSD/geoagent/microduck-lab/rlx/runs/studio/swing/swing-e2e-20260907-v3/swing.safetensors
Source SHA-256	de820bf3f448374d12ab4dc3673f65dcc5d37f7c821455e093cd1a3c76b077fe
Teacher-assisted	yes
Initial policy role	pretrained baseline, not a null control
Comparison seed	501
PPO raw-return change vs initialization	890.5029

This run continues a pretrained policy. Its initial skill is not credited to PPO. A nonzero parameter change proves optimization occurred, not that it improved the initializer. Compare the baseline and final physical outcomes and raw returns before attributing any benefit to PPO.

ONNX, controls, and source identity

Check	Value
Policy SHA-256	d0599d4453fe80bcac4e92cb61d858dae2c86310090885c2a5ecb07ed2bbc916
ONNX input	batch x 61
ONNX output	batch x 14
ONNX contract	PASS
Native/ONNX max absolute error	1.132e-06
Parameter L2 change	0.4898
Negative controls failed as required	yes
Audit scope	Nominal local simulation only; no hardware certification. No assisted Swing starts.

API workflow

Action	Accepted	Phase	Exit code	Finished
train	yes	succeeded	0	2026-09-08T01:25:13.226Z
eval	yes	succeeded	0	2026-09-08T01:25:25.071Z
render	yes	succeeded	0	2026-09-08T01:25:46.250Z

Action	Accepted	Phase	Exit code	Finished
export	yes	succeeded	0	2026-09-08T01:25:51.887Z

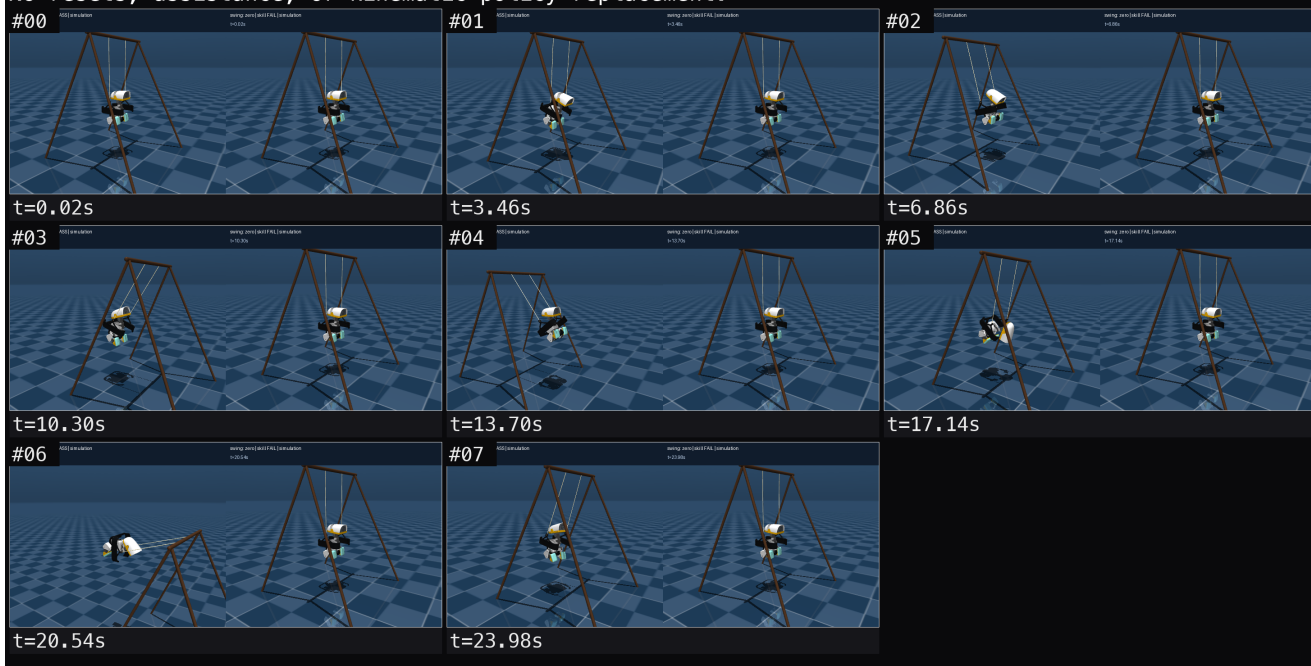
Video and graph evidence

Video validation	Value
Validation status	PASS
API bytes equal local render	yes
HTTP byte range	yes
Browser playback	{"currentTime": 12, "duration": 24, "error": null, "height": 360, "width": 640}

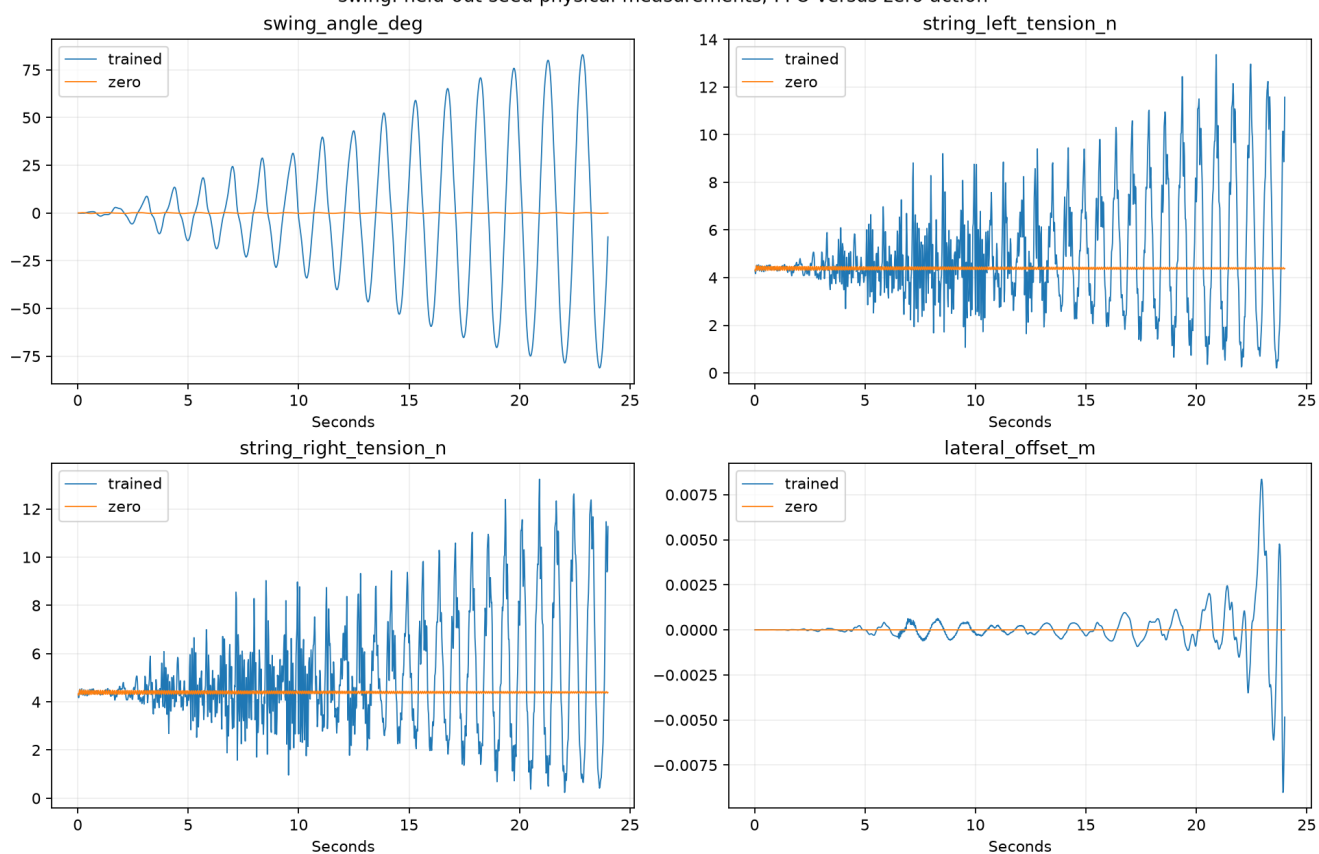
File	Codec	Dimensions	FPS	Frames	Seconds	SHA-256
api-video.mp4	h264	640 x 360	25/1	600	24.000000	dee5b2805efcd8339f9e2389c78fbc6bcd4d095e4caa693d2f2e872bdfefeb54ea
comparison.mp4	h264	960 x 412	25/1	600	24.000000	7e0ee67139bf014e8f513d7fe4c4c325c4a84bd68f7d15ec59760106310cac64

- [comparison_sheet.png](#)
- [physical-tracking.png](#)
- [reward-learning.png](#)
- [ppo-losses.png](#)
- [browser-video.png](#)
- [studio-evaluation.png](#)
- [comparison.mp4](#)
- [api-video.mp4](#)
- [audit.json](#)
- [video-validation.json](#)
- [swing-v3-api.json](#)

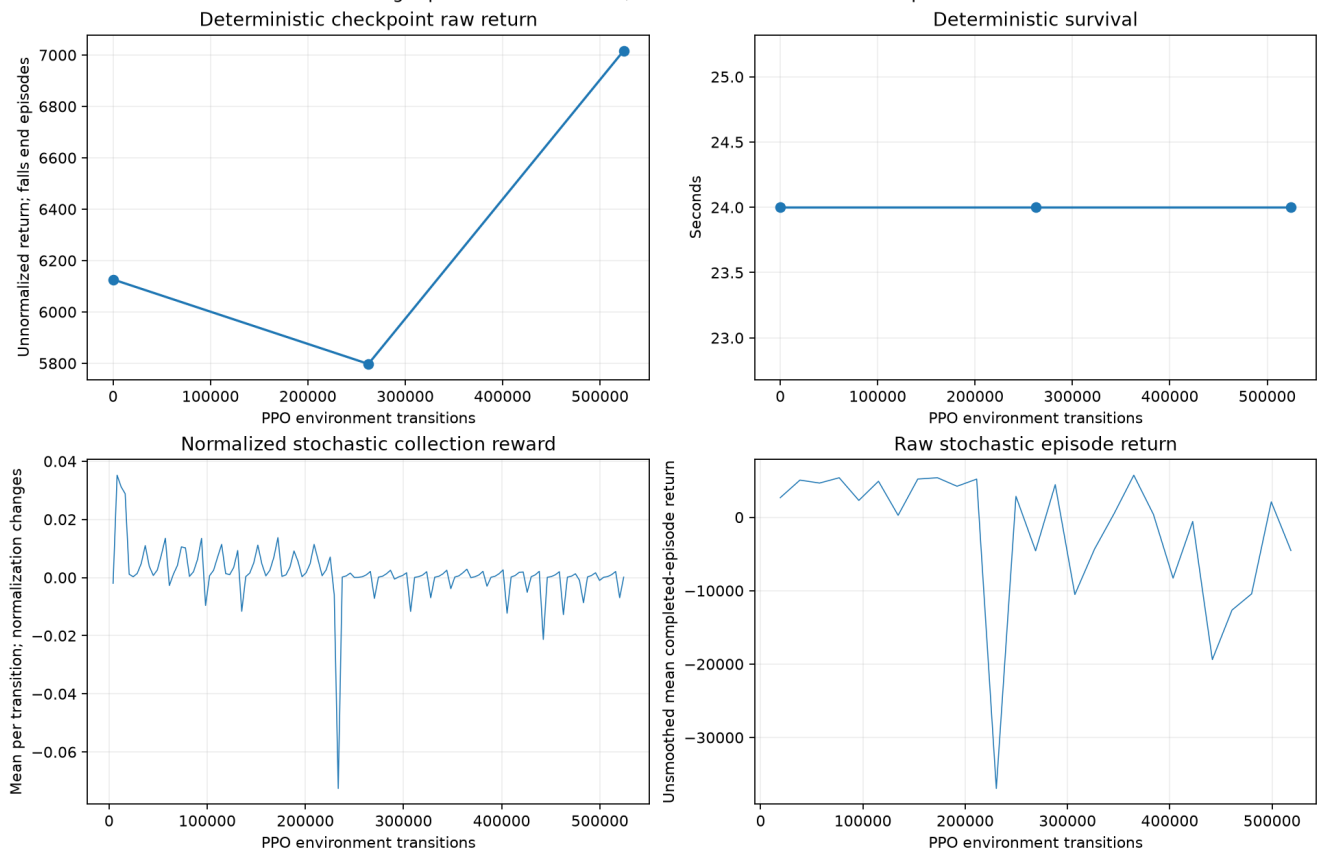
swing: trained ONNX versus zero action
No resets, assistance, or kinematic policy replacement.



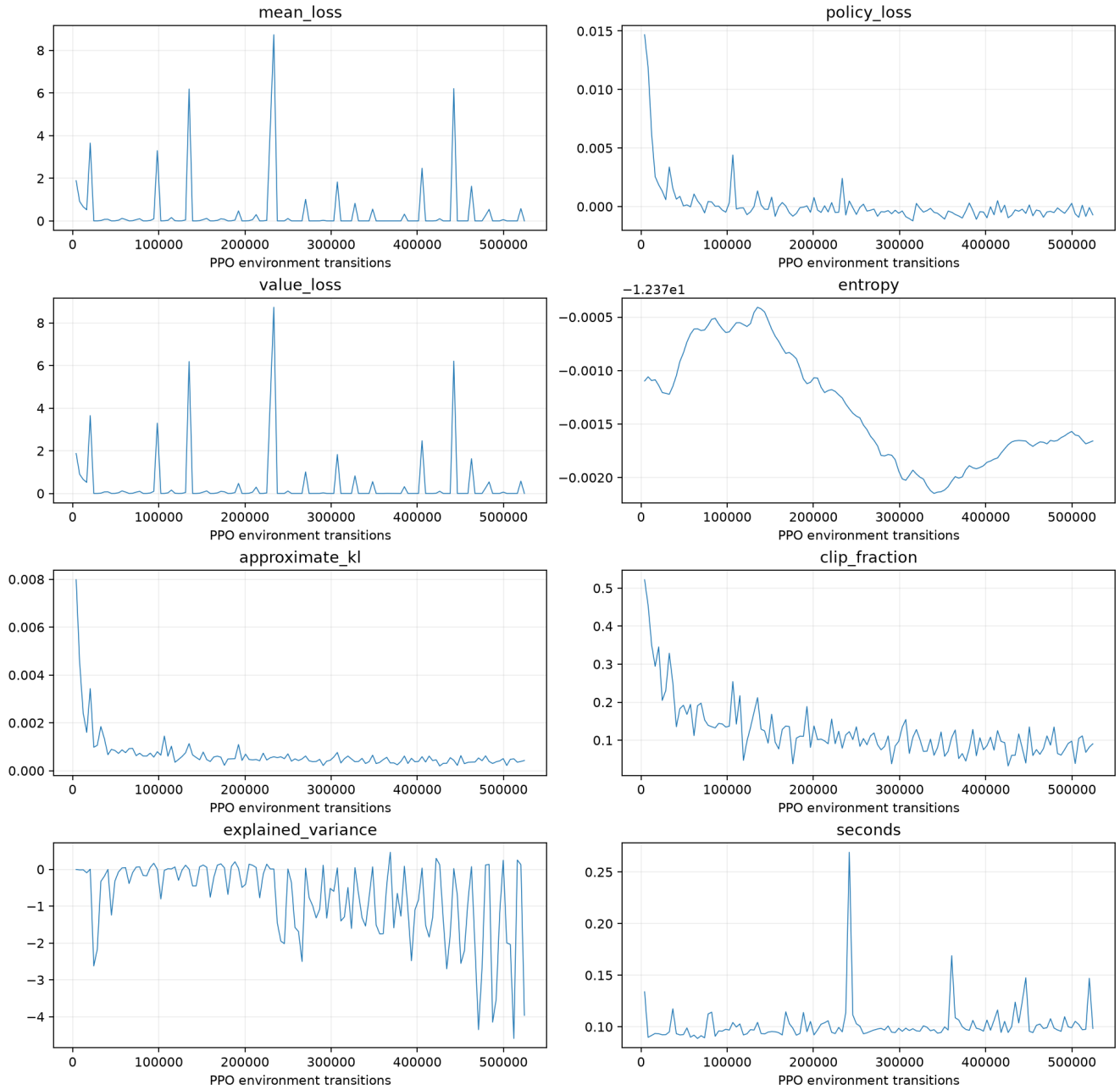
swing: held-out seed physical measurements, PPO versus zero action

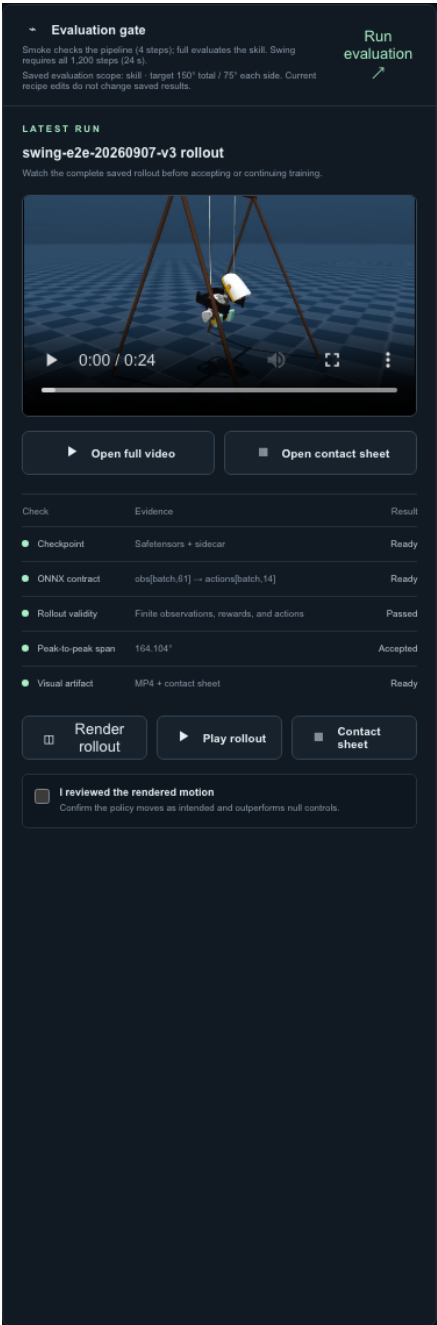


swing: optimization evidence, not a substitute for skill acceptance



Actual PPO minibatch aggregates; full history including startup spikes





Source hashes

Source/artifact	Bytes	SHA-256
swing.json	756	37c683b8146aec303e7ae63c9d328846717a2e4781f8fee0a37250c77530fe0a
swing-v3.json	756	37c683b8146aec303e7ae63c9d328846717a2e4781f8fee0a37250c77530fe0a
audit.json	52,563	e571d732c0f3b7ff19bdcd8c12efbfebda2b49d53533a6e0ad4725334cb5c3a4
video-validation.json	6,379	62ff5ca8c66d556cf3e1d2178b59da339f5330bae1b3e67cf0d1b13796838ba2
swing-v3-api.json	282,887	b64ac543dd98fd30ff7bf855b0a88dcbaaa439e8183c9665833eee6ffcabded7
comparison_sheet.png	1,492,424	8b5c97fab468760e2b903212802d9e717d11eb0a5ed2ab7f6bd063c4a0cbff71
physical-tracking.png	359,762	932448909624d357c2895170b756c23f0d2b91ddd15f72b2ae39a2f35bd4d09d

Source/artifact	Bytes	SHA-256
reward-learning.png	203,453	c9577b684f4f2478a6eeb873f26478a43e57fe002e3bb596c86aca9ebd975520
ppo-losses.png	337,856	2a5906835b76b9e842895b11ef0bcc14c9a71565138837150e1d4d4b647459d9
browser-video.png	183,643	9e463c627bc09c0359742df6521f27d24780f43658d4d63eb1b03b68df629084
studio-evaluation.png	101,087	b59406d1b25ca84380514e687a997c45492a7f7b418a112b4f65884c8e1087df
comparison.mp4	2,275,553	7e0ee67139bf014e8f513d7fe4c4c325c4a84bd68f7d15ec59760106310cac64
api-video.mp4	2,155,746	dee5b2805efcd8339f9e2389c78fbc6bcd4d095e4caa693d2f2e872bdfef54ea

Reproduction commands

Run all stages from the repository root with the Studio server at <http://127.0.0.1:63317>. Replace YYYYMMDD-HHMMSS consistently with a fresh timestamp; base and continuation runs must remain distinct.

Swing v3 repeats bootstrap PPO with a 524,288-transition budget, not a continuation of the failed v2 final policy. The v2 524k checkpoint passed while its 1M final failed; this shorter schedule still requires a new held-out audit and is not a predeclared pass.

This recipe requires a teacher-assisted initializer. Reproduce the BC/Dagger stage first; it uses privileged state only to create training labels, never as an inference input to the exported actor. Preserve the initializer's own verdict.

```
rlx/.venv-microduck/bin/python rlx/scripts/bootstrap_swing_e2e.py --source docs/remaining-scenarios-e2e/
  recipes/swing-teacher.json --output /tmp/swing-bootstrap-reproduction
mkdir -p rlx/runs/studio/swing/swing-reproduction-YYYYMMDD-HHMMSS
test ! -e rlx/runs/studio/swing/swing-reproduction-YYYYMMDD-HHMMSS/swing.safetensors
cp /tmp/swing-bootstrap-reproduction/swing.safetensors /tmp/swing-bootstrap-reproduction/
  swing.safetensors.json rlx/runs/studio/swing/swing-reproduction-YYYYMMDD-HHMMSS/
```

Run from the repository root after starting the Studio server at <http://127.0.0.1:63317>:

```
node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
  swing --recipe-json docs/remaining-scenarios-e2e/recipes/swing.json --run swing-reproduction-
  YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/swing-reproduction-YYYYMMDD-HHMMSS-
  api.json --timeout-seconds 7200
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json docs/remaining-scenarios-e2e/
  recipes/swing.json --run rlx/runs/studio/swing/swing-reproduction-YYYYMMDD-HHMMSS --output rlx/
  artifacts/scenarios-e2e-20260907/swing-reproduction-YYYYMMDD-HHMMSS-audit --seeds 501 502 503 504
  505 --render
node duck-viewer/scripts/verify-rlx-video.mjs --recipe-json docs/remaining-scenarios-e2e/recipes/
  swing.json --run rlx/runs/studio/swing/swing-reproduction-YYYYMMDD-HHMMSS --output rlx/artifacts/
  scenarios-e2e-20260907/swing-reproduction-YYYYMMDD-HHMMSS-audit --base-url http://127.0.0.1:63317
  --playwright-package "${PLAYWRIGHT_PACKAGE:-/Users/ghu/community/package.json}"
```

The audit command exits nonzero when the strict skill gate fails. That nonzero exit is expected evidence for a failed attempt, not permission to rewrite the result as successful.

8. Historical Running: running-v1

Recorded skill status: NOT RUN. Not a selected final policy.

Full raw evaluations, historical curves, and videos remain linked below:

- [running-v1-api.json](#)

9. Historical Running: running-v2

Recorded skill status: FAIL. Not a selected final policy.

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	2,930
Optimizer minibatches	46,880
Observed transitions	6,000,640
Requested transitions	6,000,640
Monotonic update steps	yes
Training evidence gate	PASS
Maximum approximate KL	1.2958
Last-100 mean_loss	-0.0089
Last-100 policy_loss	-0.0097
Last-100 value_loss	0.0008
Last-100 entropy	0.2441
Last-100 approximate_kl	0.1124
Last-100 clip_fraction	0.5453
Last-100 explained_variance	0.8010

Recorded per-episode failures:

- **seed 101, episode 0:** aerial fraction below running target; command-directed displacement below target
- **seed 102, episode 0:** aerial fraction below running target; command-directed displacement below target
- **seed 103, episode 0:** aerial fraction below running target; command-directed displacement below target
- **seed 104, episode 0:** aerial fraction below running target; command-directed displacement below target
- **seed 105, episode 0:** aerial fraction below running target; command-directed displacement below target

Full raw evaluations, historical curves, and videos remain linked below:

- [comparison_sheet.png](#)
- [physical-tracking.png](#)
- [reward-learning.png](#)
- [ppo-losses.png](#)

- [browser-video.png](#)
- [comparison.mp4](#)
- [api-video.mp4](#)
- [audit.json](#)
- [video-validation.json](#)
- [running-v2-api.json](#)
- [running-v2-api.log](#)

10. Historical Running: running-v3

Recorded skill status: FAIL. Not a selected final policy.

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	2,930
Optimizer minibatches	46,880
Observed transitions	6,000,640
Requested transitions	6,000,640
Monotonic update steps	yes
Training evidence gate	PASS
Maximum approximate KL	1.2383
Last-100 mean_loss	-0.0051
Last-100 policy_loss	-0.0073
Last-100 value_loss	0.0022
Last-100 entropy	2.0482
Last-100 approximate_kl	0.1647
Last-100 clip_fraction	0.5357
Last-100 explained_variance	0.7433

Recorded per-episode failures:

- **seed 301, episode 0:** command-directed displacement below target
- **seed 302, episode 0:** none
- **seed 303, episode 0:** command-directed displacement below target
- **seed 304, episode 0:** command-directed displacement below target
- **seed 305, episode 0:** none

Full raw evaluations, historical curves, and videos remain linked below:

- [comparison_sheet.png](#)
- [physical-tracking.png](#)
- [reward-learning.png](#)
- [ppo-losses.png](#)

- [browser-video.png](#)
- [comparison.mp4](#)
- [api-video.mp4](#)
- [audit.json](#)
- [video-validation.json](#)
- [running-v3-api.json](#)
- [running-v3-api.log](#)

11. Historical Stilt Walking: stilts-v1

Recorded skill status: TRAIN FAILED. Not a selected final policy.

Training stopped at 1,980,416 transitions: RuntimeError: PPO gradients contains non-finite values: gradient tensors=critic.layers.0.weight, critic.layers.0.bias, critic.layers.2.weight, critic.layers.2.bias; observations[-7.21088..10; nonfinite=0], actions[-1.39449..3.84149; nonfinite=0], old_log_probabilities[-11.5659..8.13558; nonfinite=0], old_values[-1.64562..3.35344; nonfinite=0], advantages[-1.87452..1.62534; nonfinite=0], returns[-0.430656..3.33284; nonfinite=0]

Full raw evaluations, historical curves, and videos remain linked below:

- [stilts-v1-api.json](#)

12. Historical Stilt Walking: stilts-v2

Recorded skill status: FAIL. Not a selected final policy.

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	2,930
Optimizer minibatches	46,880
Observed transitions	6,000,640
Requested transitions	6,000,640
Monotonic update steps	yes
Training evidence gate	PASS
Maximum approximate KL	0.7728
Last-100 mean_loss	-0.0045
Last-100 policy_loss	-0.0100
Last-100 value_loss	0.0055
Last-100 entropy	1.9163
Last-100 approximate_kl	0.0814
Last-100 clip_fraction	0.4561

Training evidence	Value
Last-100 explained_variance	0.0249

Recorded per-episode failures:

- **seed 101, episode 0:** command-directed displacement below target
- **seed 102, episode 0:** command-directed displacement below target
- **seed 103, episode 0:** command-directed displacement below target
- **seed 104, episode 0:** command-directed displacement below target
- **seed 105, episode 0:** none

Full raw evaluations, historical curves, and videos remain linked below:

- [comparison_sheet.png](#)
- [physical-tracking.png](#)
- [reward-learning.png](#)
- [ppo-losses.png](#)
- [browser-video.png](#)
- [comparison.mp4](#)
- [api-video.mp4](#)
- [audit.json](#)
- [video-validation.json](#)
- [stilts-v2-api.json](#)

13. Historical Swing: swing-v1

Recorded skill status: FAIL. Not a selected final policy.

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	512
Optimizer minibatches	6,144
Observed transitions	1,048,576
Requested transitions	1,048,576
Monotonic update steps	yes
Training evidence gate	PASS
Maximum approximate KL	0.0664
Last-100 mean_loss	-0.0711
Last-100 policy_loss	-0.0074
Last-100 value_loss	9.801e-06
Last-100 entropy	12.7553
Last-100 approximate_kl	0.0034
Last-100 clip_fraction	0.1767

Training evidence	Value
Last-100 explained_variance	-4.0719

Recorded per-episode failures:

- **seed 101, episode 0:** bidirectional span below target
- **seed 102, episode 0:** bidirectional span below target
- **seed 103, episode 0:** bidirectional span below target
- **seed 104, episode 0:** bidirectional span below target
- **seed 105, episode 0:** bidirectional span below target

Full raw evaluations, historical curves, and videos remain linked below:

- [comparison_sheet.png](#)
- [physical-tracking.png](#)
- [reward-learning.png](#)
- [ppo-losses.png](#)
- [browser-video.png](#)
- [comparison.mp4](#)
- [api-video.mp4](#)
- [audit.json](#)
- [video-validation.json](#)
- [swing-v1-api.json](#)

14. Historical Swing: swing-v2

Recorded skill status: FAIL. Not a selected final policy.

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	256
Optimizer minibatches	2,048
Observed transitions	1,048,576
Requested transitions	1,048,576
Monotonic update steps	yes
Training evidence gate	PASS
Maximum approximate KL	0.0080
Last-100 mean_loss	0.3409
Last-100 policy_loss	-0.0005
Last-100 value_loss	0.3415
Last-100 entropy	-12.3716
Last-100 approximate_kl	0.0004
Last-100 clip_fraction	0.0724

Training evidence	Value
Last-100 explained_variance	-2.5277

Recorded per-episode failures:

- **seed 301, episode 0:** invalid swing geometry; one or both strings lost positive spring tension
- **seed 302, episode 0:** invalid swing geometry; one or both strings lost positive spring tension
- **seed 303, episode 0:** invalid swing geometry; one or both strings lost positive spring tension
- **seed 304, episode 0:** invalid swing geometry; one or both strings lost positive spring tension
- **seed 305, episode 0:** invalid swing geometry; one or both strings lost positive spring tension

Full raw evaluations, historical curves, and videos remain linked below:

- [comparison_sheet.png](#)
- [physical-tracking.png](#)
- [reward-learning.png](#)
- [ppo-losses.png](#)
- [browser-video.png](#)
- [comparison.mp4](#)
- [api-video.mp4](#)
- [audit.json](#)
- [video-validation.json](#)
- [swing-v2-api.json](#)
- [swing-v2-api.log](#)

15. Stilt training failure and ELU diagnosis

The retained `stilts-v1` API record establishes the observed failure: PPO aborted at 1,980,416 of 4,001,792 transitions because critic-layer gradients became non-finite. The logged observations, actions, old log probabilities, old values, advantages, and returns were finite, and no final checkpoint or ONNX artifact was produced.

A separate installed-MLX activation probe supplies a plausible mechanism. The stock `mlx.nn.elu` produced finite forward outputs for large positive inputs 90 and 1000 but non-finite gradients. This is consistent with the critic-only gradient failure because actor and critic use the same hidden activation structure while receiving different objectives and gradient paths. It is **not proof of the exact failing minibatch**, which was not saved.

The local model now uses a stable ELU expression:

```
where(x >= 0, x, exp(min(x, 0)) - 1)
```

This keeps the same 512-256-128 hidden-layer topology, checkpoint tensor layout, and ELU forward behavior used by exported ONNX graphs. The PPO finite-value guards remain fail-closed: a non-finite loss, gradient, or model parameter still raises; bad updates are not silently skipped.

Probe field	Captured value
Inputs	[-1.0, 0.0, 1.0, 90.0, 1000.0]
Outputs	[-0.6321205496788025, 0.0, 1.0, 90.0, 1000.0]
Gradients	[0.3678794503211975, 1.0, 1.0, null, null]
Finite outputs	yes
Finite gradients	no
Implementation	installed mlx.nn.elu: where($x > 0$, x , $\alpha * (\exp(x) - 1)$)

Validation artifact	Result	Link
Activation test before fix	3 failed because <code>stable_elu</code> did not yet exist	activation-before.log
Activation test after fix	3 passed: eager/compiled large-input gradient boundary and forward parity	activation-after.log
Installed MLX probe	finite forward, non-finite gradients at large positive inputs	activation-probe.json

Source hashes:

Artifact	Bytes	SHA-256
activation-probe.json	375	0074fd2183fd765c729d5746162adc302e62bd84dd0755149d0dd682830fbbe5
activation-before.log	2,068	0dab9d24d3d427ed44db37ea020c3680b9cac27c1c77ed79883208389d0b04f2
activation-after.log	98	fc99c56adf2cb5702a4564ceead9e104cf6db01c3cd1a15756b5c9630e644ce6
activation_probe.py	769	c1a52763100be1cdf44f16da9def46a54964fc10c1445999d78d38066c9bd03c

16. Tracked reproduction recipes

These copies are tracked with the report so a fresh checkout does not depend on the gitignored evidence directory:

- [Running](#)
- [Running base](#)
- [Stilt Walking](#)
- [Stilt Walking base](#)
- [Swing](#)

17. Interpretation

What was tested, and what was not

This extends the Dance workflow to the Studio **Running**, **Stilt Walking**, and **Swing** recipes. The executable chain is real HTTP training, checkpoint export, deterministic skill evaluation, MuJoCo rendering, artifact download, H.264 decoding, and Chromium playback/seek. The

separate audit uses the exported ONNX actor, not a scripted motion replay. Its comparison video reconstructs the recorded physics states without resetting a fallen control. The displayed zero-action failure is retained, not replaced by a new episode.

All final tests use the nominal XML actuator model. They do not establish BAM actuator equivalence, randomized robustness, arbitrary-speed command following, or hardware readiness. The locomotion recipes use fixed, observable forward commands; the stilt result applies to the selected **2 cm, blend 0, 0.014 kg-per-foot morphology**, not every height offered by the application. Swing uses planar action restrictions as declared in the recipe, starts at zero angle and angular rate, and must retain positive tension in both compliant strings at every 50 Hz sample.

The five final audit seeds are additional reset checks, not five independently trained policies. Swing's nominal reset is effectively identical across seeds, so its repeated passes demonstrate execution reproducibility, **not five independent robustness trials**. Locomotion resets vary, but all share the stated nominal configuration. Checkpoint-development seeds were used for tuning; final audit seed lists and all failed attempts remain visible in the evidence tables.

Why the early locomotion attempts failed

Running v1 learned standing. Running v2 learned fast grounded stepping but did not satisfy the aerial-phase gate. Running v3 acquired an actual aerial gait around 0.7 m/s, but its trajectories often curved far enough to fail net intended displacement. A high body-frame speed alone was therefore insufficient. The evaluator was strengthened before final selection to measure signed progress against the intended command heading, rather than crediting accumulated travel around a circle. The acceptance floors were not reduced to accommodate these failures.

The optional `flight` term is zero by default and rewards both feet being airborne only with upright posture and command-directed speed. Its bounded form prevents stationary hopping or a falling robot from collecting full credit. Running's optional `yaw_tracking` term isolates yaw-rate error from the existing combined roll/pitch/yaw Gaussian: $\exp(-((\text{gyro_z} - \text{commanded_yaw_rate}) / 0.25)^2)$. This gives a yaw-specific signal while leaving the native gyro, gravity, joint, prior-action, and command observations unchanged. It does not expose absolute world heading to the actor or reward an unobservable heading target. The same opt-in term is supported for stilts, but the selected Stilt v3 recipe does **not** enable it.

Running: an aerial gait with verified forward progress

Running v4 resumes the 6,000,640-transition v3 actor for **2,097,152 additional PPO transitions**, with learning rate 0.0001, yaw-only weight 8, and combined angular-tracking weight reduced from 8 to 2. The forward command remains 0.75 m/s and the flight term remains 4. This changes the learning objective, not the evaluation criteria or the robot's physics.

All 16 API episodes and all five final audit seeds pass. The latter sustain **0.738–0.775 m/s, 3.848–9.013 m** signed intended displacement in twelve seconds, **35.0–39.8%** both-feet airborne samples, and **167–170** alternating-support switches, with 100% upright coverage. On seed 501, raw deterministic return under the same final reward configuration improves by about **1,683** versus the resumed initializer. The unnormalized checkpoint curve rises overall and dips slightly at the final checkpoint; this dip is retained. World-path plots still show curvature. The result establishes running and the declared progress floor, not perfect straight-line regulation or hardware-quality gait smoothness.

Stilt Walking: PPO acquisition followed by refinement

The Stilt v2 base was trained from random initialization for 6,000,640 transitions. It learned moving, alternating-foot balance on the selected physical stilt geometry, but many episodes circled and failed intended progress. Stilt v3 resumes that base for 1,048,576 additional PPO transitions and increases the existing angular-velocity tracking weight from 0.5 to 4. The inherited exploration standard deviation is loaded from the checkpoint; the recipe's initial-standard-deviation field does not reset a resumed actor.

The final API passes all 16 episodes. The separate seeds 501–505 pass all five episodes with mean forward speeds 0.230–0.242 m/s and intended displacements 1.281–2.289 m over ten seconds, without a fall. On comparison seed 501, deterministic raw return rises by about 758 under the **same final reward configuration**. This is stronger evidence of beneficial refinement than merely showing changed weights. The final gait still has measurable curvature; passing the defined progress floor does not mean perfectly straight tracking.

Swing: distinguish teacher acquisition from PPO improvement

Fresh PPO in Swing v1 did not discover the large-amplitude motion: its strict still-start span was only about 2.23 degrees. Rather than crediting assisted resets as success, a bounded controller feasibility search and behavior-cloning/DAGger initializer were used to place useful motion in the policy's reachable distribution. Privileged physical state supplies teacher labels during data collection only. The exported actor still receives the ordinary **61 inputs and produces 14 actions**; no teacher controller is called at evaluation time. The bootstrap has `ppo_steps=0` and must not be described as PPO learning.

The selected stage-three initializer already passes the Swing skill gate. PPO therefore refines an acquired skill; it does not acquire Swing from scratch in the successful branch. Observation statistics are frozen for this continuation so changing normalization cannot silently alter the competent baseline. The critic and actor are still optimized, rewards remain normalized with separately updated statistics, and all normal PPO finite-value guards remain active.

Swing v2 showed why a final checkpoint cannot be accepted solely because training finished: its 524,288-step checkpoint passed, but continued optimization to 1,048,576 steps caused geometry violations and one sampled loss of string tension. Swing v3 repeats the initialization and PPO recipe with the shorter **524,288-transition budget** selected from that development evidence. Its API result has approximately **162.32 degrees** of symmetric span, 100% valid geometry and tension, and at least **0.209 N** sampled spring tension. This is a model-selection decision with an explicitly retained failed longer run, not a smoothed graph or relaxed threshold. Alignment approaches the 0.05 limit, so the pass is nominal and has limited margin.

The independent ONNX audit reproduces the pass and increases seed-501 raw return from **6,126.39 to 7,016.89**, about **890.50** or **14.5%**, compared with the saved initial checkpoint. This supports a beneficial PPO refinement of a teacher-acquired skill. It does not justify relabeling the bootstrap's behavior cloning as PPO or hiding the subsequent failed longer continuation.

How to read the reward and loss figures

Each figure comes from saved JSONL telemetry or actual deterministic checkpoint rollouts. No reward or loss series is invented, made monotonic, or truncated to remove startup spikes. The reward figure deliberately separates deterministic unnormalized checkpoint return, deterministic

survival, normalized stochastic collection reward, and raw stochastic completed-episode return. Those are different measurements: normalizer scale evolves, training actions are sampled, and occasional exploratory falls shorten episodes. A downward normalized training curve is not by itself evidence that the exported deterministic policy got worse.

For example, Stilt v3's deterministic checkpoint return increases while its normalized stochastic reward decreases during part of training. Its occasional raw training-return drops coincide with shortened exploratory episodes; the independent deterministic gait is the acceptance measurement. PPO policy loss is a batch-relative clipped surrogate, not a supervised prediction-error curve. The reports retain value loss, approximate KL, clip fraction, entropy, explained variance, and update duration so a small scalar loss cannot conceal an unstable or uninformative critic. In particular, poor or negative explained variance in Swing is disclosed rather than presented as healthy value-function convergence.

Numerical repair and evidence integrity

Stilt v1 aborted on nonfinite critic gradients. A separate installed-MLX probe reproduced finite ELU forward outputs but NaN gradients for large positive inputs. Bounding the inactive exponential branch preserves the ELU function and ONNX layout while eliminating that reproduced gradient-overflow mechanism. Regression tests cover eager and compiled gradients, forward parity, checkpoint compatibility, and the unchanged observation/action interface. The exact failed training minibatch was not saved, so the probe establishes a consistent failure mechanism rather than an exact replay of the original failure.

The evidence retains unsuccessful runs and distinguishes a completed pipeline from a learned skill. Null controls must fail, exported ONNX must match the checkpoint, final episodes must meet their physical gates, and video validation must establish actual downloadable and playable files. The native tests, viewer tests, lint/build results, and known broader-suite failures are reported separately; optional GPU-stack collection failures are not recast as passing CPU coverage. Nothing in this report certifies deployment to a real robot.

The browser test also visits the actual Studio page, selects each of the three experiments, loads the final run by name, verifies its **Accepted** skill verdict, and plays the embedded video at the expected 12/10/24-second duration. Those screenshots complement the direct artifact-download and standalone Chromium playback checks. No UI test clicks Train, changes the completed policy, or treats the disconnected live lab as a policy failure.

That test caught the rollout player's two-column minimum widths overflowing the narrow evaluation card. The player now stacks inside the card, its copy refers to the saved rollout rather than always claiming 24 seconds, and the browser check asserts zero horizontal overflow and scroll offset for all three final runs.

Implementation map and reproduction boundaries

File or area	Change and purpose
rlx/rlx/models/microduck.py	Stable ELU activation; same model topology, checkpoint keys, and ONNX function.
rlx/rlx/environments/locomotion_evaluation.py	Fail-closed per-episode locomotion acceptance, including net intended displacement and contact/aerial evidence.

File or area	Change and purpose
<code>rlx/rlx/environments/microduck_recipes.py</code>	Physical locomotion metrics, optional fixed command, bounded flight/yaw shaping, and existing morphology/physics reuse.
<code>rlx/rlx/environments/microduck.py</code>	Optional observation-statistics freeze for a competent warm start; reward statistics remain separate.
<code>rlx/examples/ppo_microduck_studio.py</code>	Native PPO training/evaluation integration, continuation provenance, normalization settings, and saved skill assessments.
<code>duck-viewer/lib/rlx-job.ts</code> and <code>experiments.ts</code>	Recipe/API plumbing, correct full evaluation horizons, optional reward weights, and retention of only matching saved evaluations.
<code>duck-viewer/lib/evaluation.ts</code>	Explicit skill status required for all built-in scenarios; pipeline success is not task acceptance.
<code>duck-viewer/scripts/rlx-dance-api-e2e.mjs</code>	Shared four-scenario HTTP driver; preserves failed-skill render/export evidence but exits nonzero.
<code>rlx/scripts/audit_scenarios.py</code>	Independent deterministic ONNX/control evaluation, all-checkpoint curves, PPO telemetry, hashes, and no-reset physics video.
<code>duck-viewer/scripts/verify-rlx-video.mjs</code>	Hash-linked artifact bytes, range requests, H.264 frame counts, duration, and Chromium playback/seek.
<code>rlx/scripts/bootstrap_swing_e2e.py</code>	Bounded, explicitly non-PPO teacher BC/DAGger initializer; teacher labels never become privileged inference inputs.
<code>docs/remaining-scenarios-e2e</code>	Tracked selected/base/teacher recipes, guarded reproduction, report generator, interpretation, and PDF build.

The existing RLX PPO optimizer and rollout infrastructure are reused, not replaced with an animation or a teacher controller at inference. No new dependencies were installed. The successful locomotion continuations require their base checkpoints; the reproduction commands train those bases and copy both safetensors and normalization sidecars only after checking successful training. A base skill failure is retained and allowed only as the specifically recognized failure of a completed training run. Reusing a stale or failed-training checkpoint is not allowed. Swing likewise requires the disclosed bootstrap before the PPO recipe.

The generated artifacts live in the gitignored `rlx/artifacts/scenarios-e2e-20260907` and `rlx/runs/studio` directories. They are present locally and linked from this report, but a fresh clone does not automatically contain the MP4s, checkpoints, or raw traces. Retain those directories with the report when sharing a complete evidence package, or rerun the tracked reproduction recipes. The unchanged Dance evidence remains a separate report.

Primary references for interpreting PPO

These references explain the optimization and evaluation concepts, not the outcome of these particular runs. This project executes RLX PPO; the SB3 implementation is cited for metric definitions and scaling cautions, not as the engine used here.

1. Schulman et al., *Proximal Policy Optimization Algorithms* (2017), original clipped-surrogate method: <https://arxiv.org/abs/1707.06347>.
2. OpenAI Spinning Up, *Proximal Policy Optimization*, stochastic exploration, policy/value objectives and approximate-KL diagnostics: <https://spinningup.openai.com/en/latest/algorithms/ppo.html>.

3. Stable-Baselines3 2.6.0 PPO source documentation, policy/value/entropy losses, approximate KL, clipping fraction and reward-scaling caveat: https://stable-baselines3.readthedocs.io/en/v2.6.0/_modules/stable_baselines3/pppo/ppo.html.
4. Stable-Baselines3, *Reinforcement Learning Tips and Tricks*, separate evaluation of stochastic training policies: https://stable-baselines3.readthedocs.io/en/master/guide/rl_tips.html.

18. Final conclusion

Report result: PASS.

All three selected scenarios have matching successful API jobs, passing independent audits, passing video validation, and current artifact hashes. This remains simulation-only evidence under the stated XML conditions; Swing uses a teacher-assisted initializer before PPO.